

Real-World Applications & Global Change Projections

Modeling the distribution of biodiversity under global climate change

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Future climate change projections

Unravelling biogeographic patterns and future changes

Global Change Biology

WILEY



RESEARCH ARTICLE OPEN ACCESS

Global Cold-Water Coral Biodiversity Redistribution Under Projected Climate Change

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Keywords: biodiversity | bioregionalization | climate change | cold-water corals | range shifts | species distribution modelling | species richness

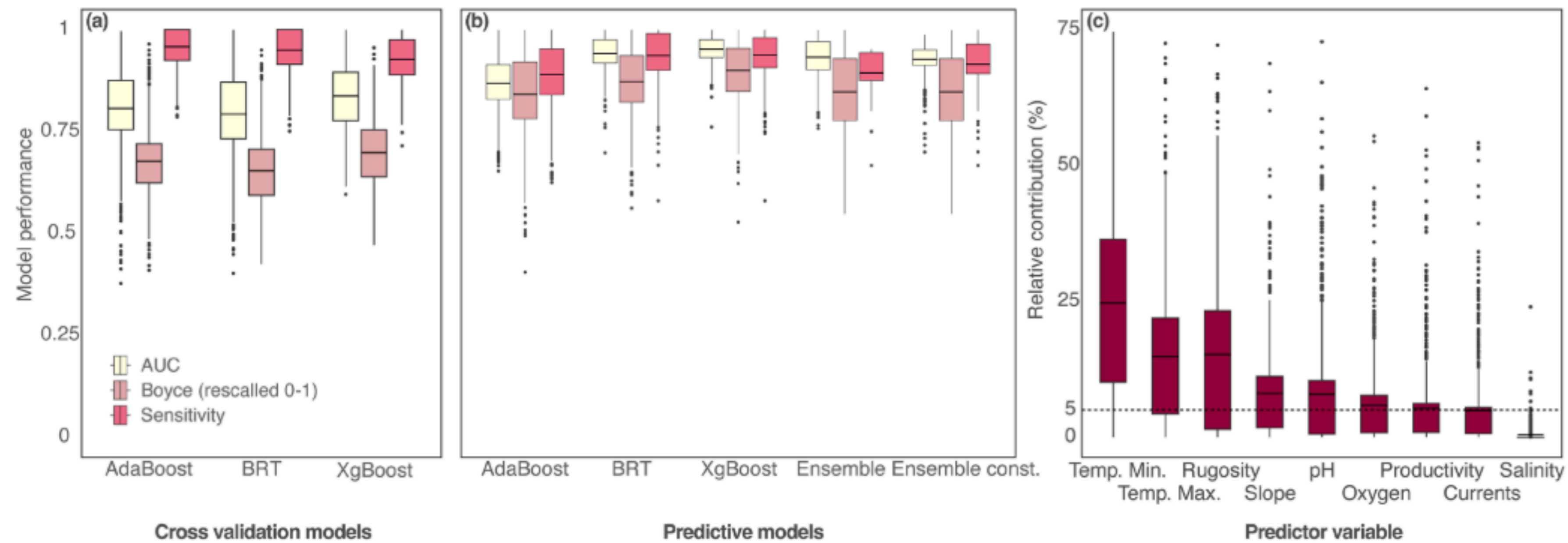
ABSTRACT

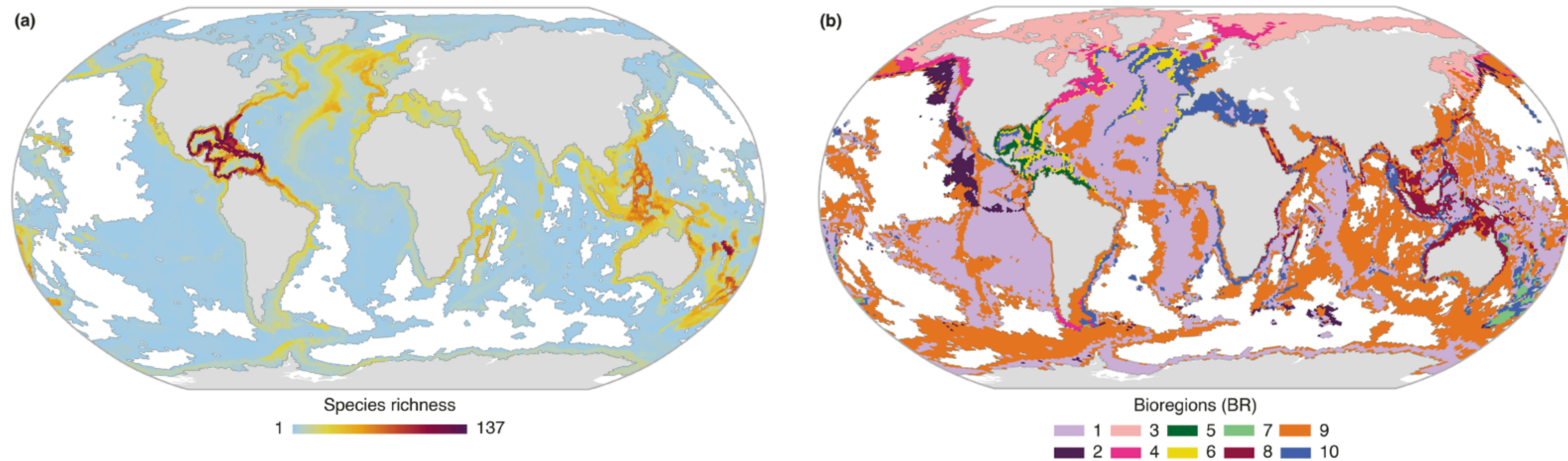


Rationale: Cold-water corals (CWCs) are **key ecosystem-structuring species** across the world's oceans, **yet their global distribution, diversity patterns, and vulnerability to climate change remain poorly understood.**

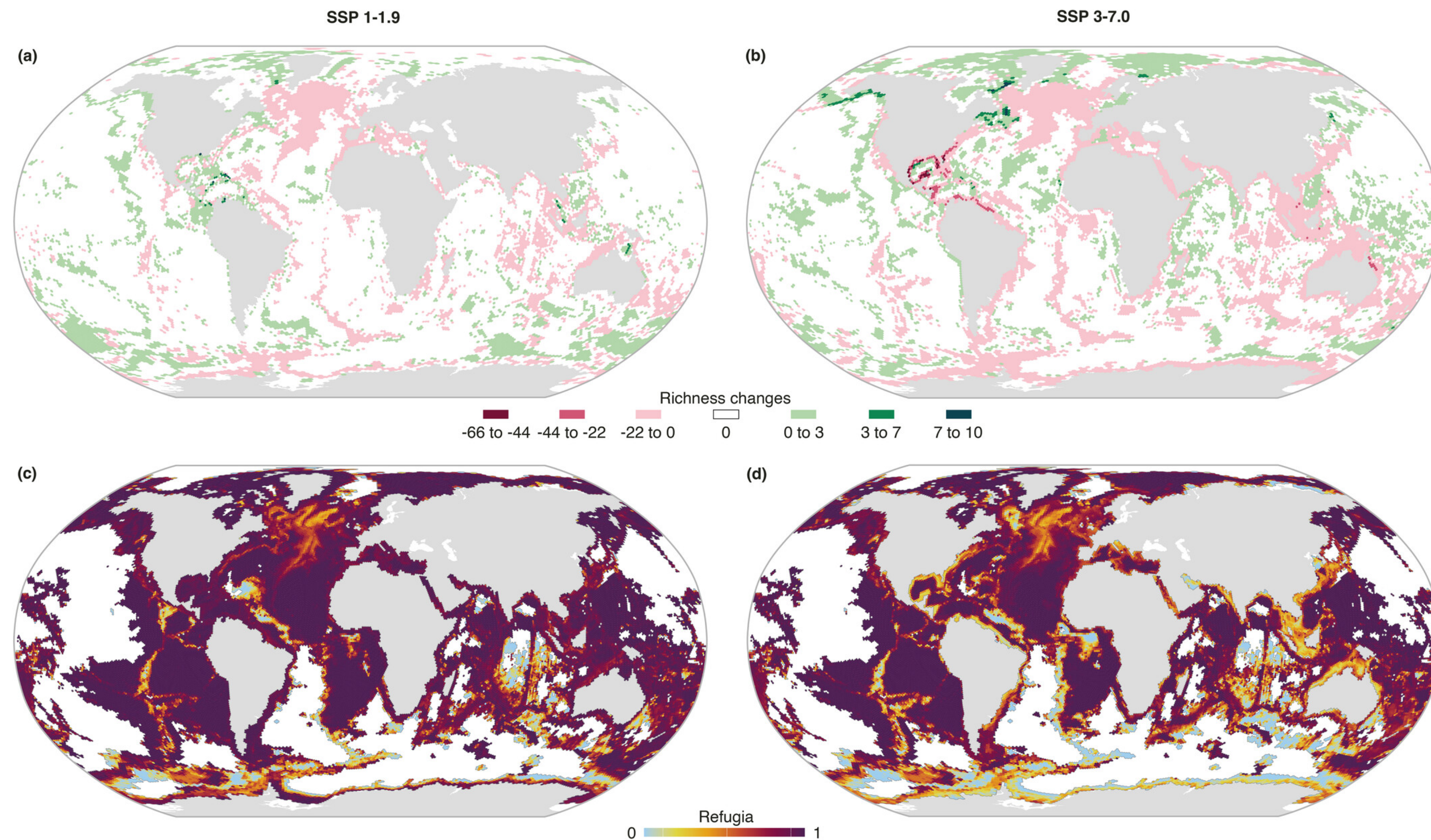
Goal: Delineate the global biogeography of CWCs and assessed how their biodiversity patterns may shift under future climate change scenarios.

Methods: Ensemble of machine-learning models predicted the distributions of 741 CWC species under present-day conditions and forecasted changes in species richness, community composition, and climate refugia under SSP1-1.9 and SSP3-7.0.





Major biodiversity hotspots in the Gulf of Mexico and the Caribbean Sea, and delineated ten distinct bioregions, each with varying species richness, depth distribution patterns, and generally low levels of endemicity.

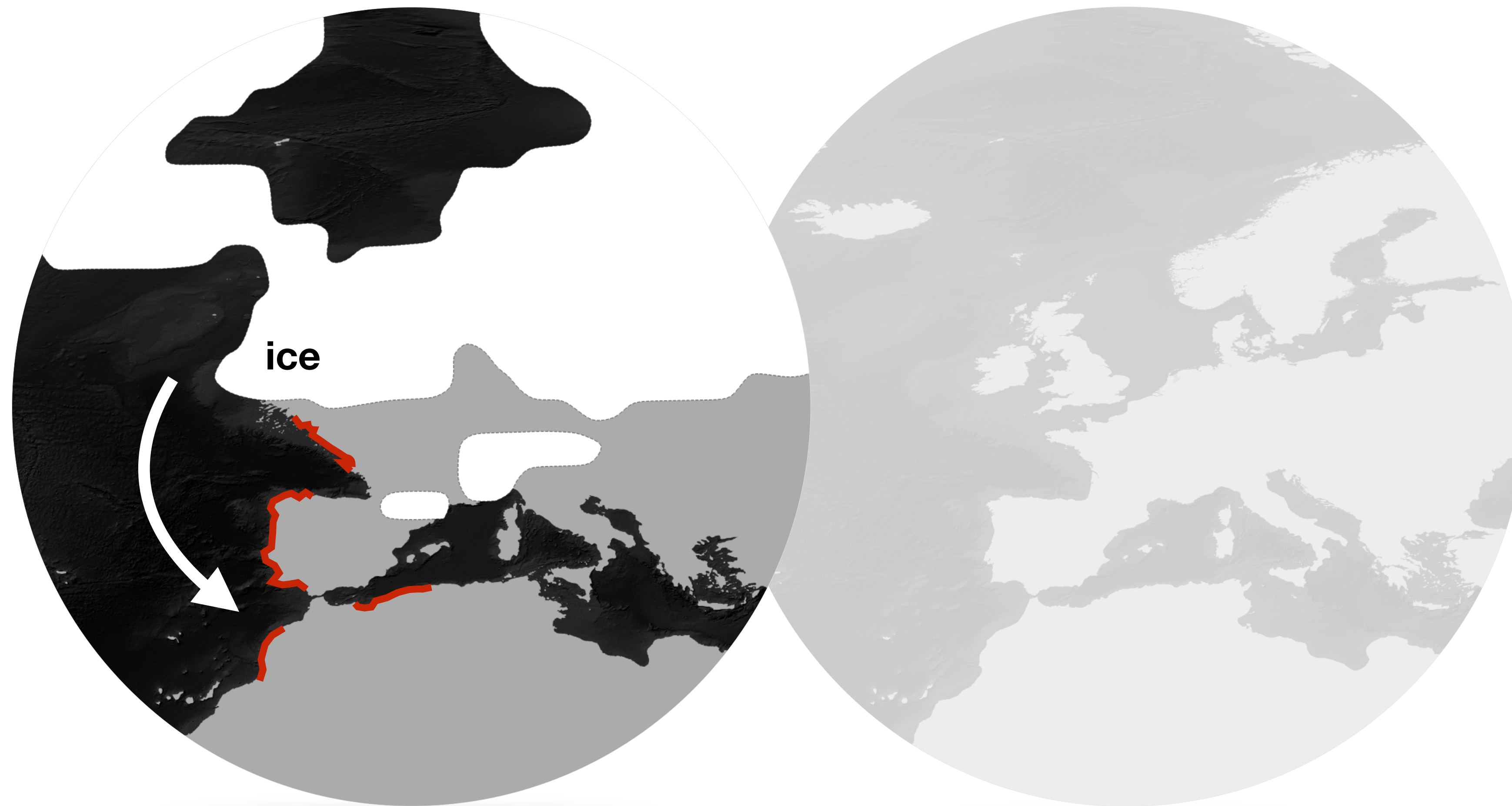


Forecasts showed **poleward shifts in species distributions**, particularly under high--emission scenarios, resulting in **biodiversity losses**, highlighting the threat of climate change to CWC biodiversity and the need for urgent climate action, aligned with the Paris Agreement. Also global and **widespread conservation opportunities (e.g., refugia)**.

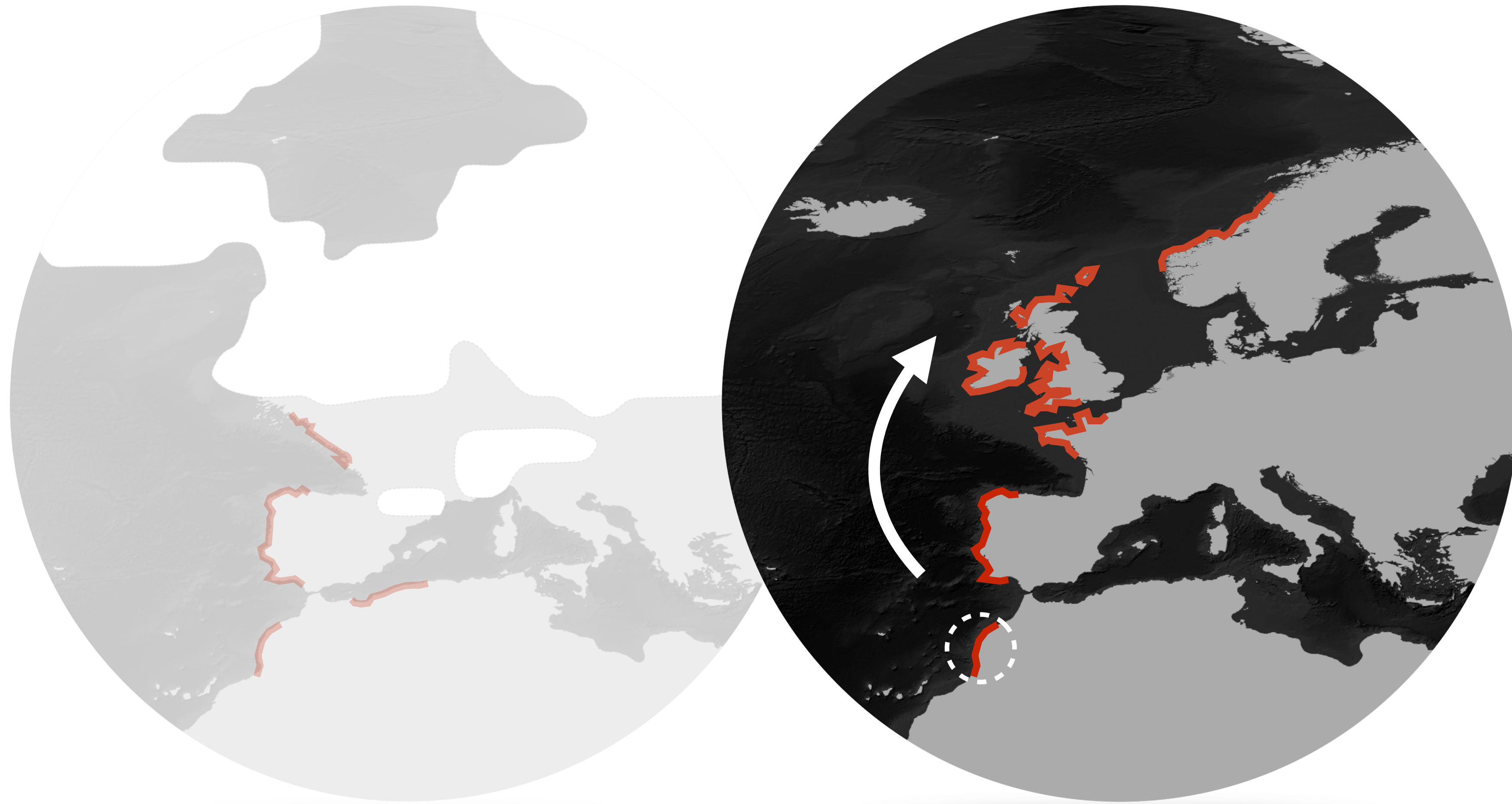
Hindcasting to Past Climates to Explain Genetic Diversity

The paradigm of past range shifts structuring the distribution of intra-specific biodiversity is well established, but it lacks replication.

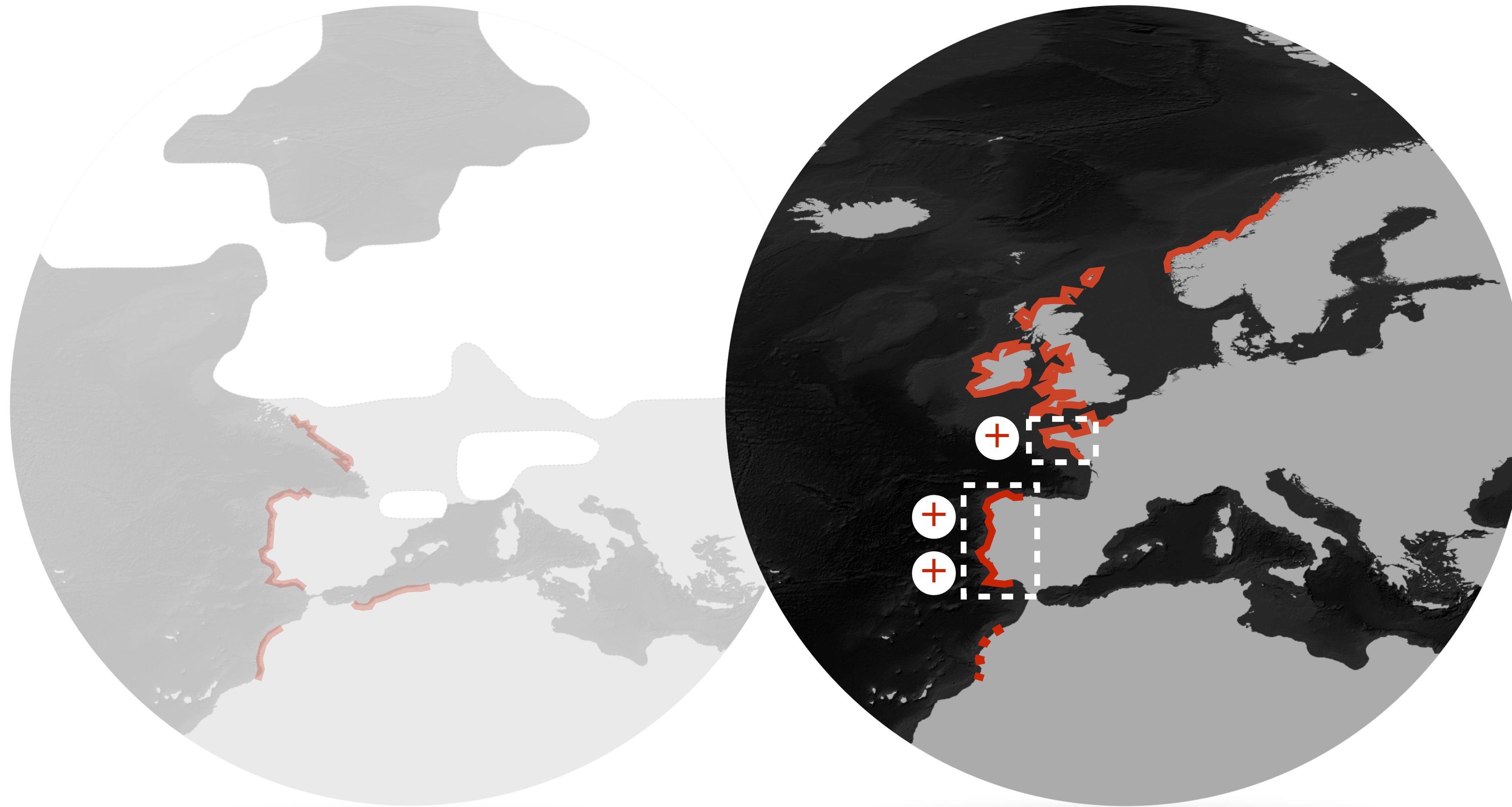
Studies addressing the topic mostly in the NE Atlantic, and **missing biological models exposed to** comparable conditions in **independent geographic regions.**



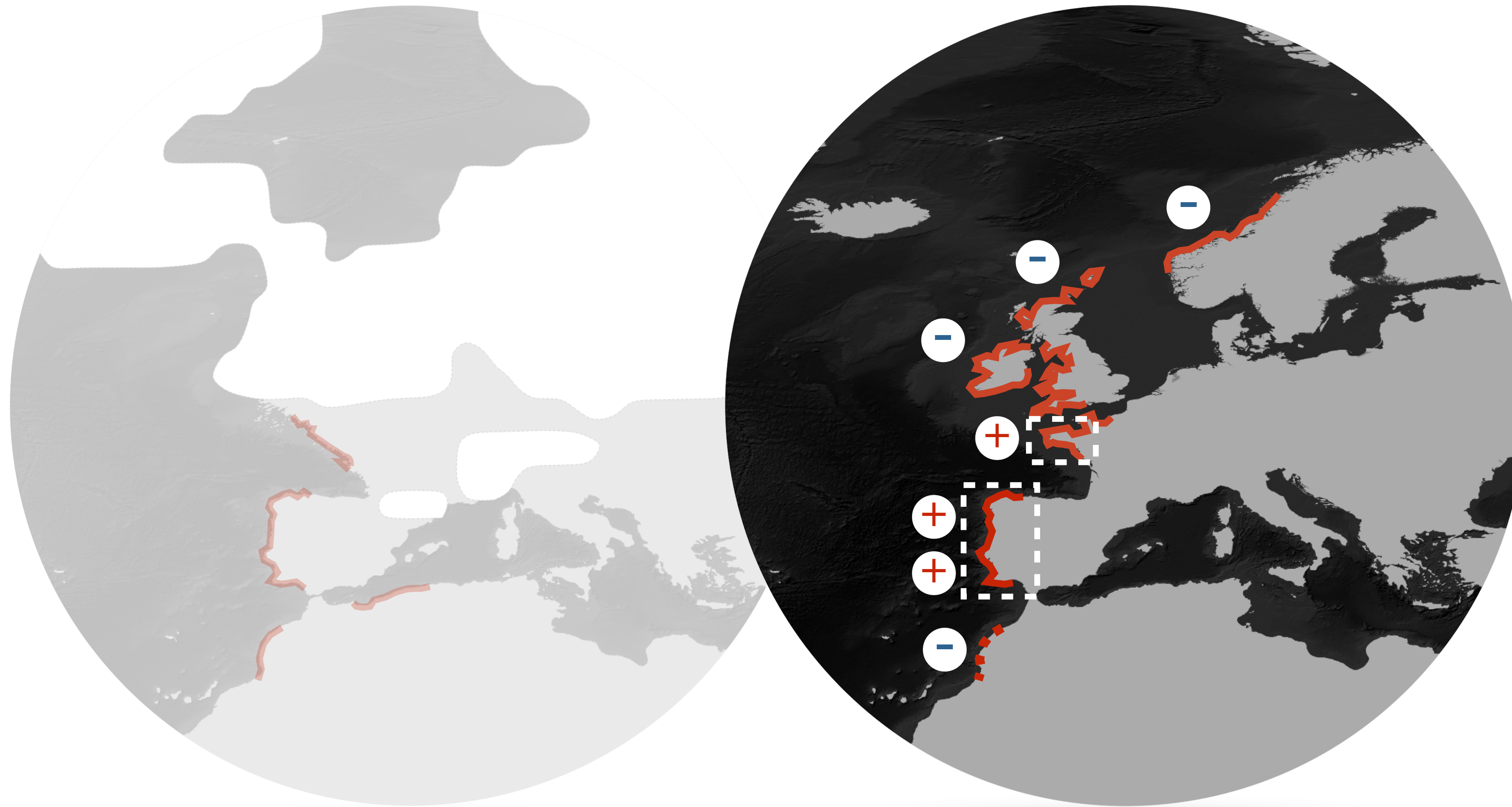
During glacial periods (e.g., LGM), the advance of massive ice sheets and the colder climate led species to contract their high latitude distributions.



During interglacials, the warmer conditions allowed species to (re)colonise higher latitudes. Extinctions might have occurred at low latitude range margins.

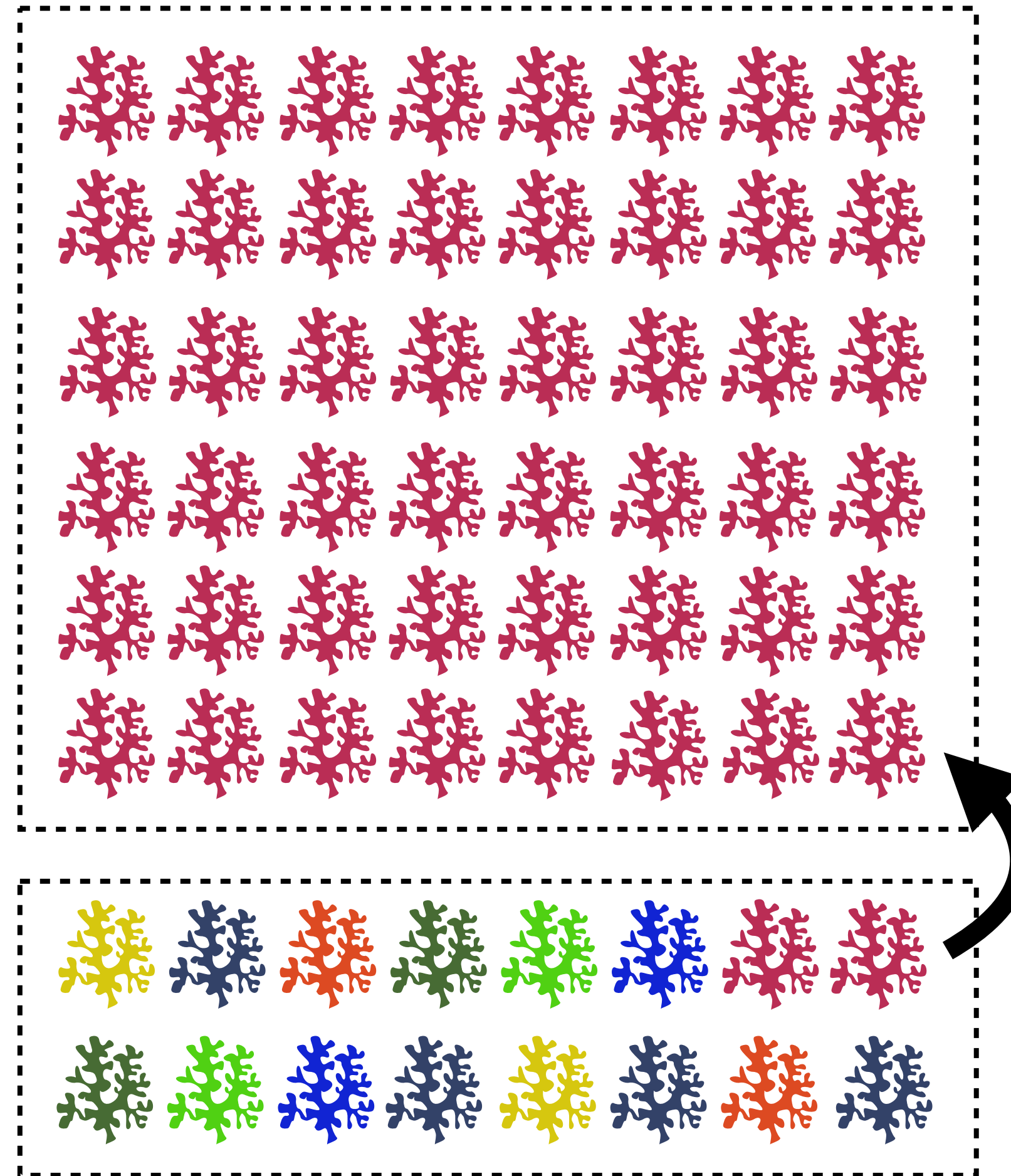


Range shifts structured regions of **long-term persistence (i.e., climatic refugia)** where populations had time to accumulate higher and unique genetic diversity.



Where local extinctions or expansions occurred, lower genetic diversity is expected, due to bottlenecks on effective population sizes or founder effects.

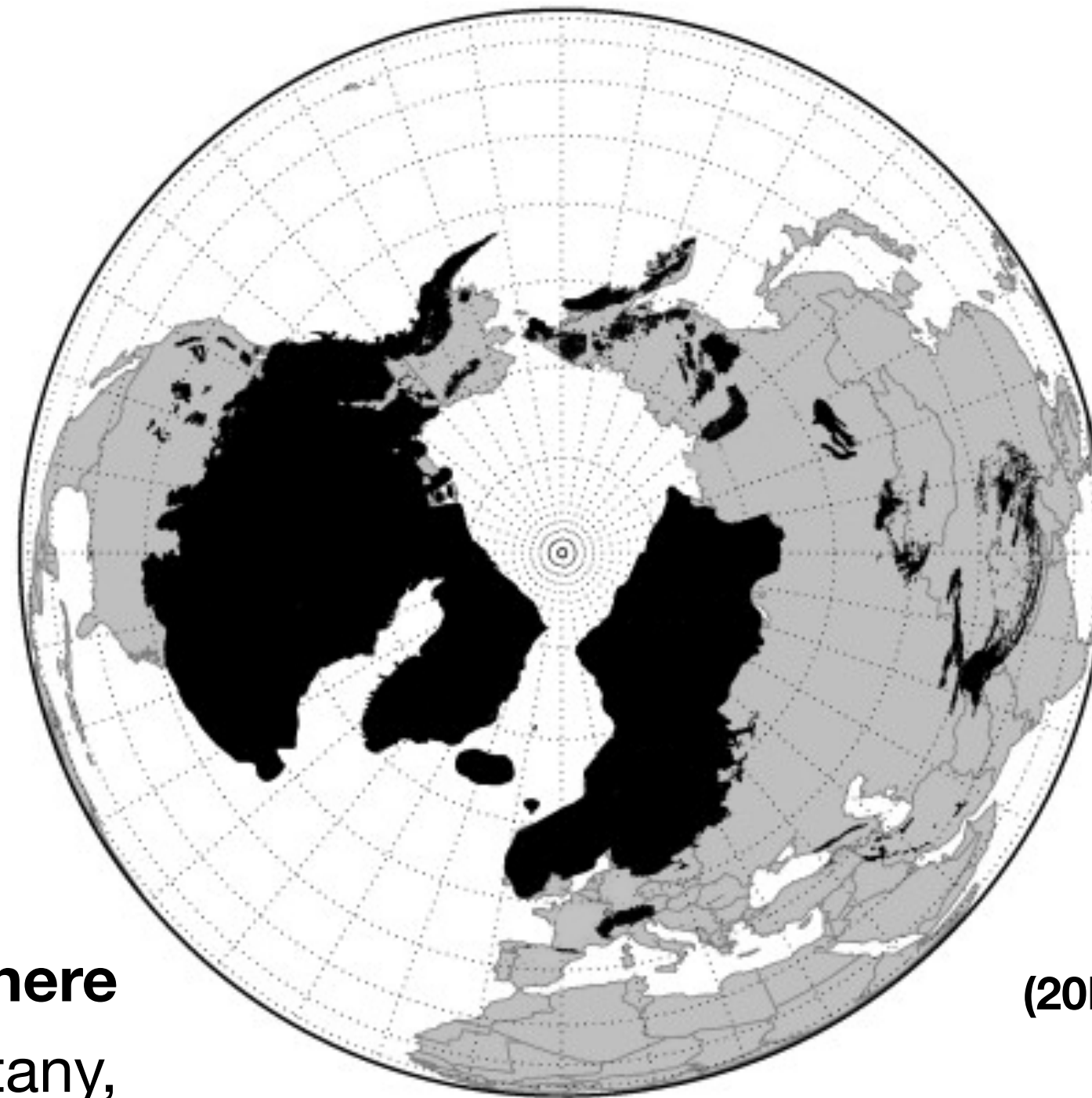
Expansion front (e.g., Post-LGM)
Low and homogeneous diversity



Founder effects **tend to leave reduced diversity** because colonizations tend to be performed by the **spread of few individuals / alleles**.

The paradigm of past range shifts structuring the distribution of intra-specific biodiversity is well established, but it lacks replication.

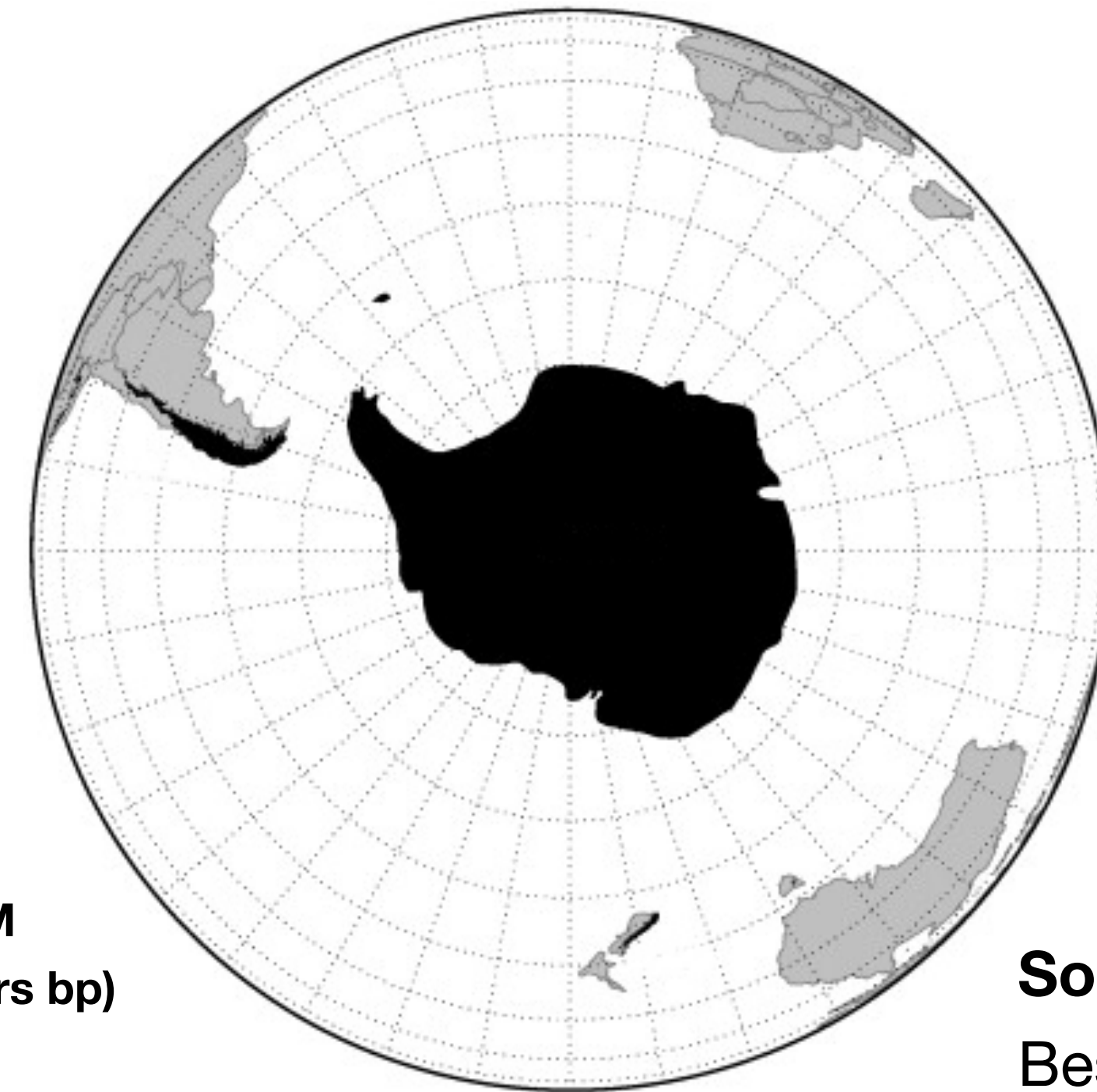
Studies addressing the topic mostly in the NE Atlantic, and **missing biological models exposed to** comparable conditions in **independent geographic regions.**



Northern hemisphere

Ice advances down to Brittany, Maine and Vancouver Island

LGM
(20k years bp)

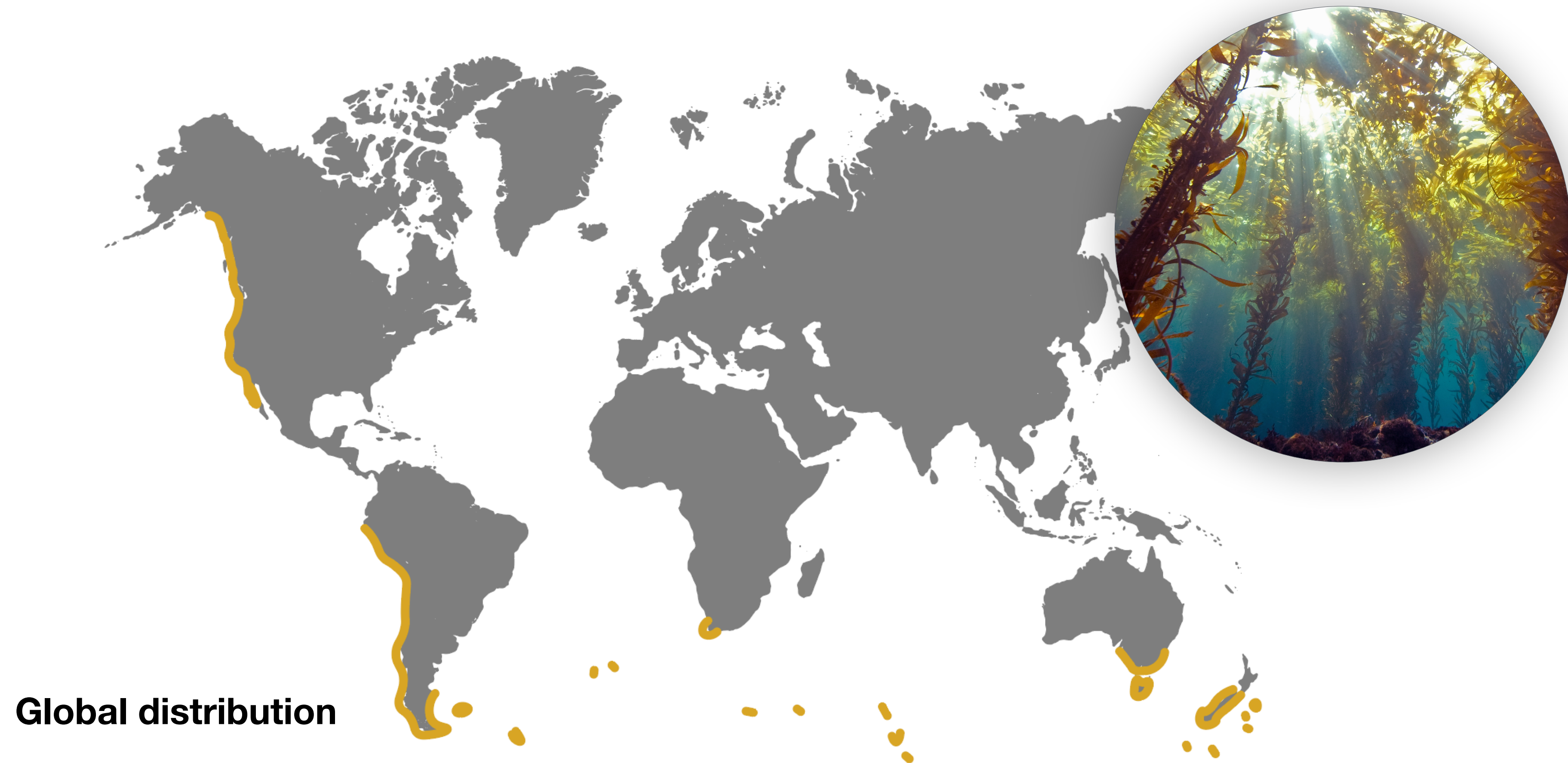


Southern hemisphere

Besides Antarctica, only Patagonia was affected

During glacial periods (e.g., LGM), the **higher latitudes of the two hemispheres provided very distinct conditions** for evolution.

** such contrasting conditions might have caused differences in the genetic diversity drivers between hemispheres*



The **giant kelp *Macrocystis pyrifera*** is an ideal model to explore the topic considering its **broad antitropical distribution** with **two independent replicate conditions** exposed to extreme glacial conditions.

Genetic diversity estimated across the entire species distribution.

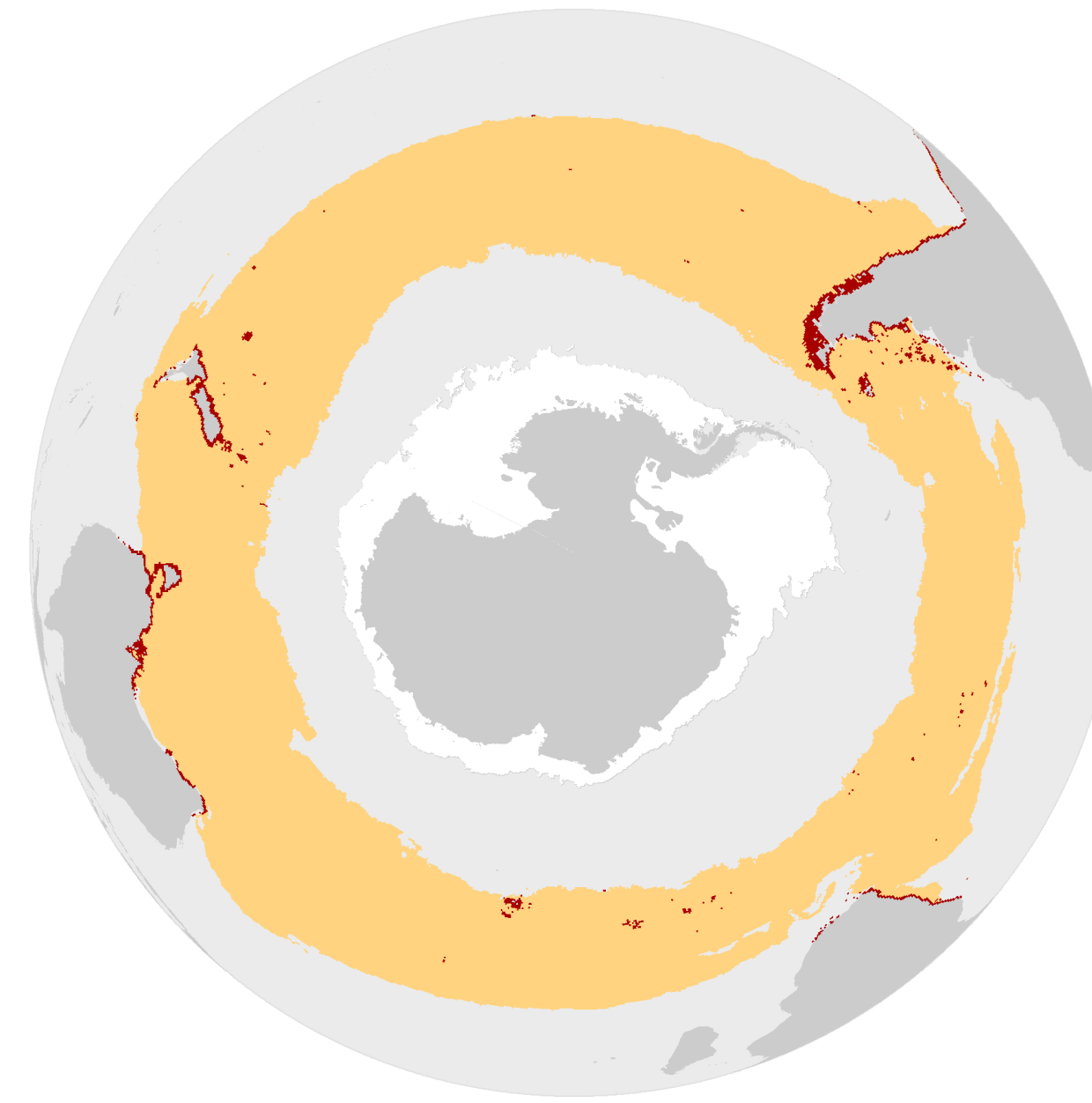
(115 sites from Alaska to the Southern Ocean)

Genetic diversity levels **compared to SDM predicting range shifts** (extinctions and expansions) **and climatic refugia** (persistence) **from the LGM to the present.**

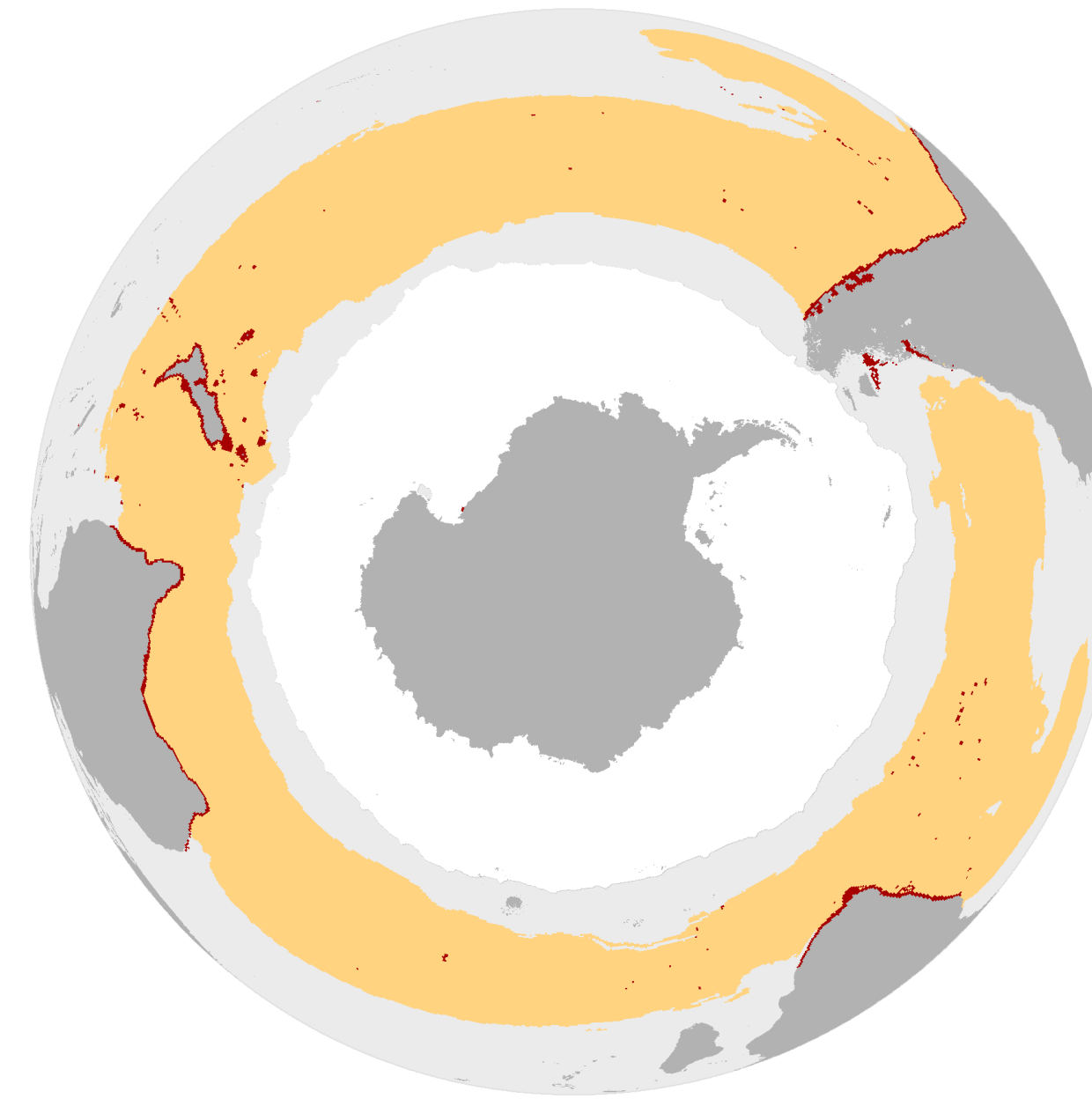
Pacific Ocean

Southern Ocean

Present-day distribution



Past distribution [LGM]

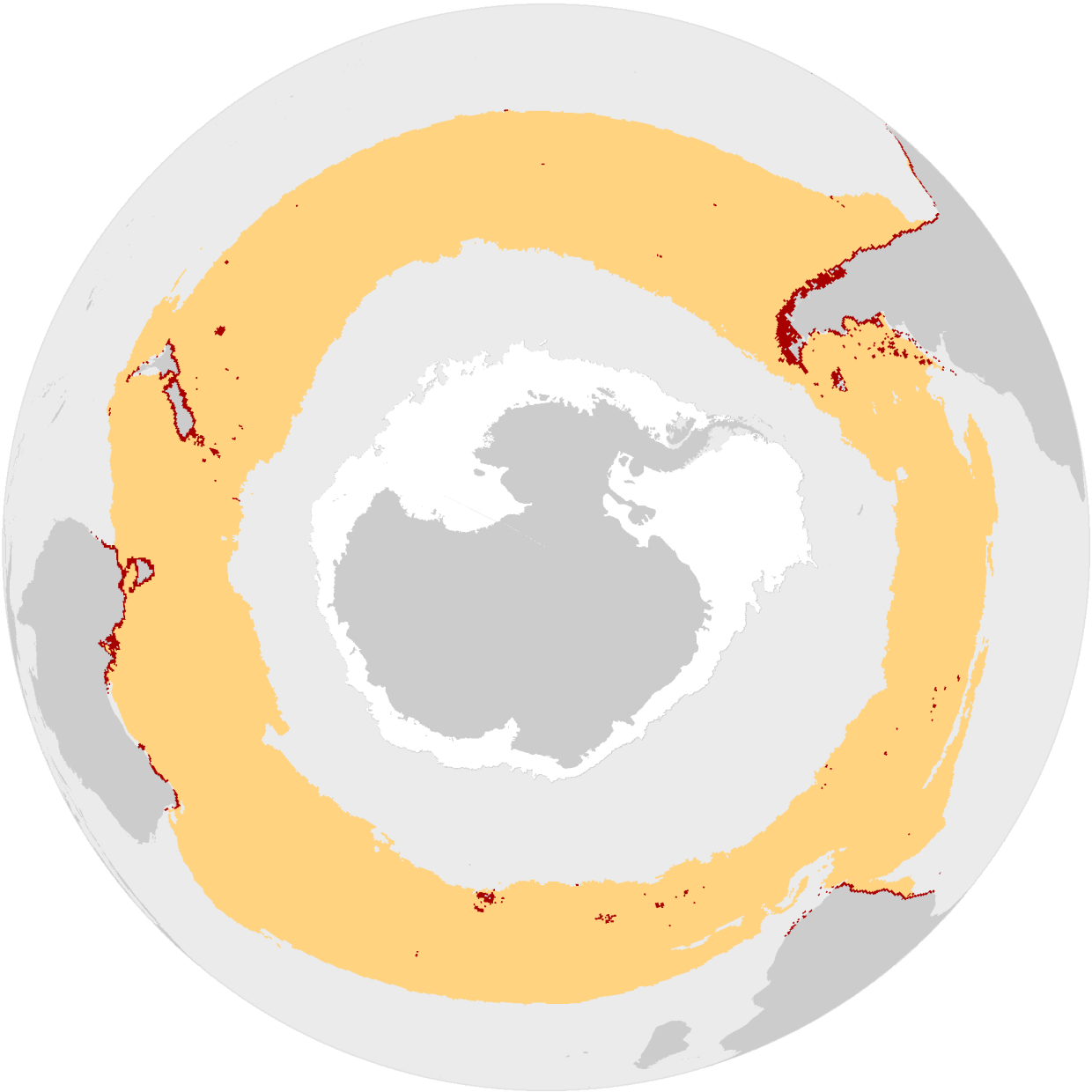


SDM predictions
● suitable habitats

Pacific Ocean

Southern Ocean

Present-day distribution



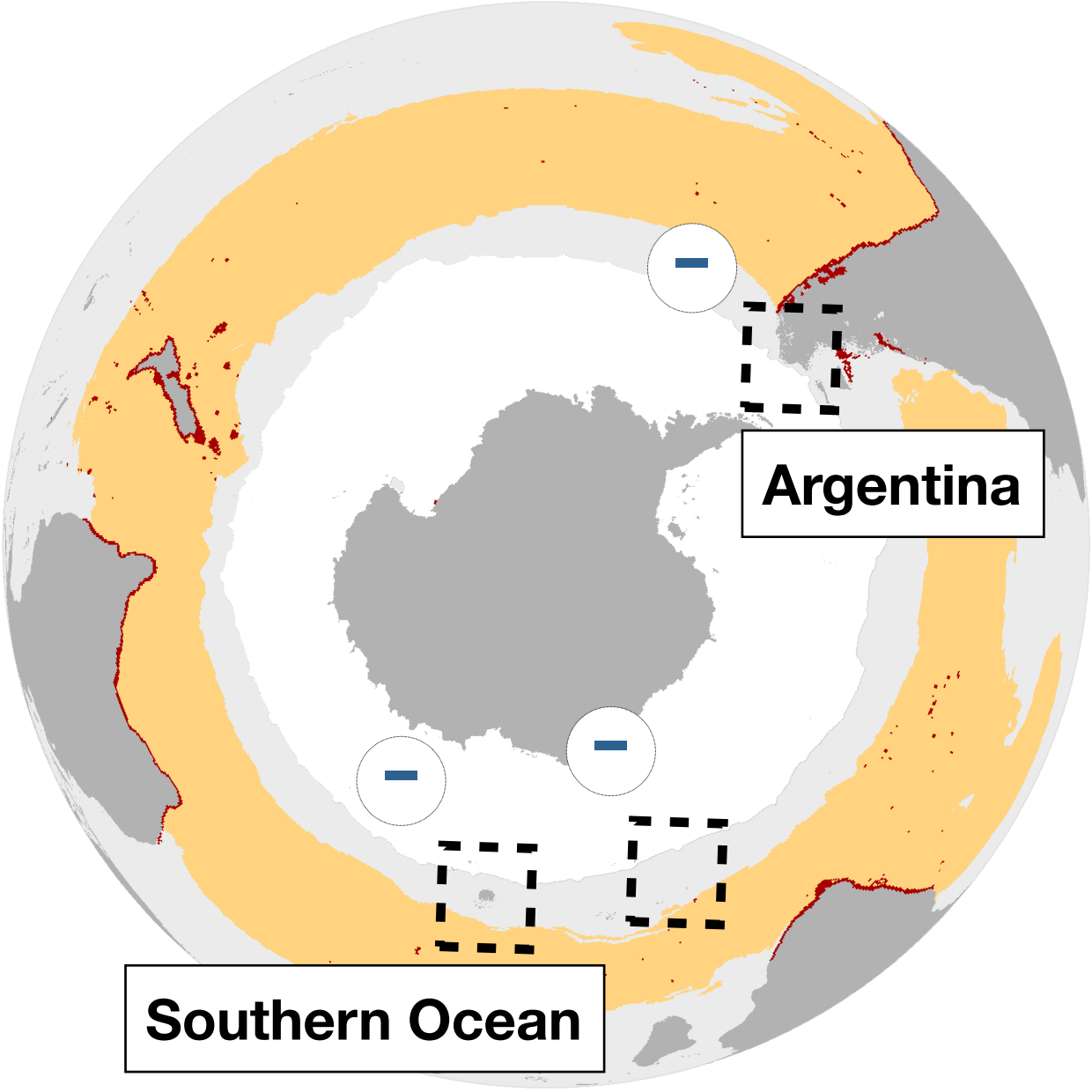
Alaska to Canada

Past distribution [LGM]



Argentina

Southern Ocean

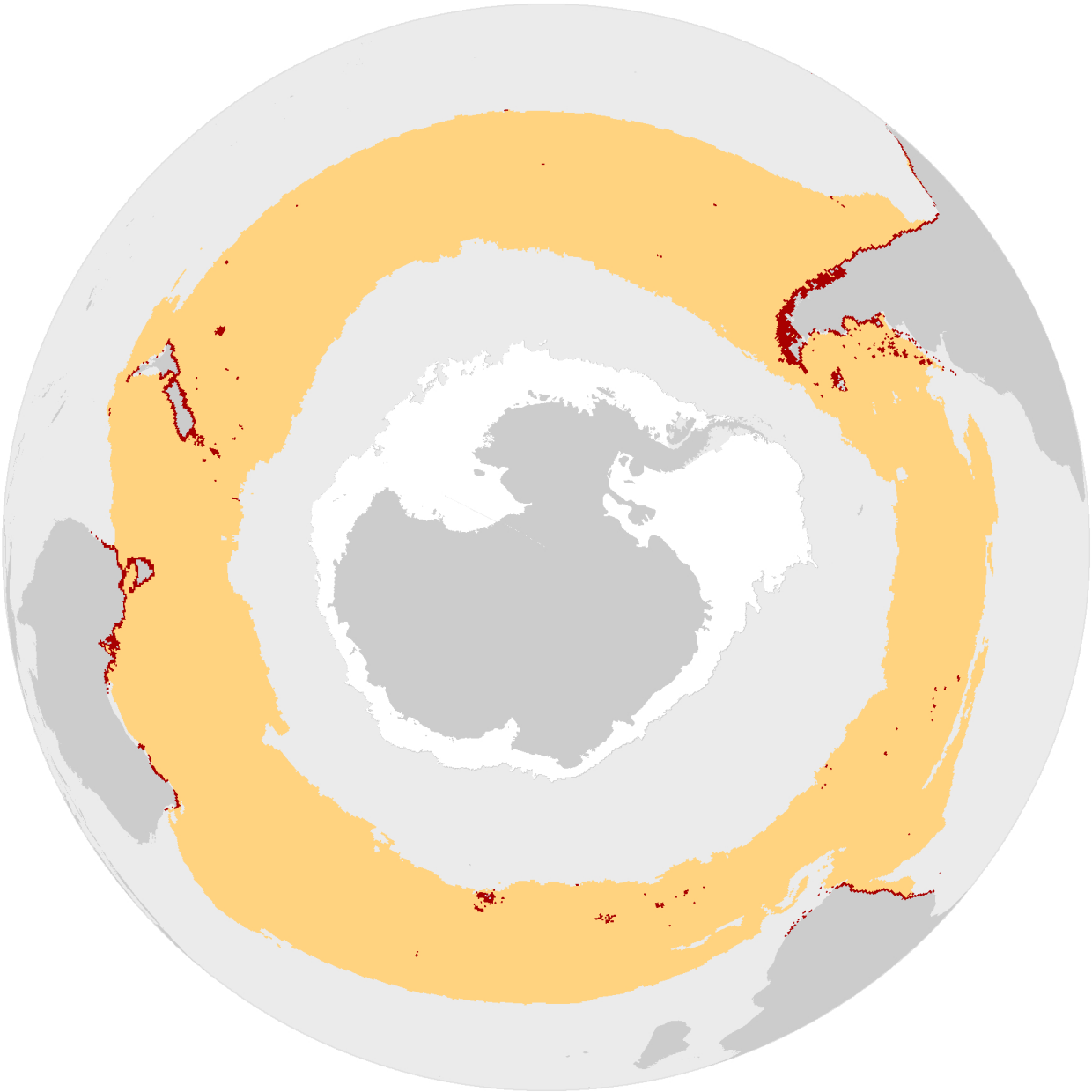


During LGM period
Broad **extinctions at higher latitudes**, leading to **40% range contraction**, globally.

Present-day distribution

Pacific Ocean

Southern Ocean



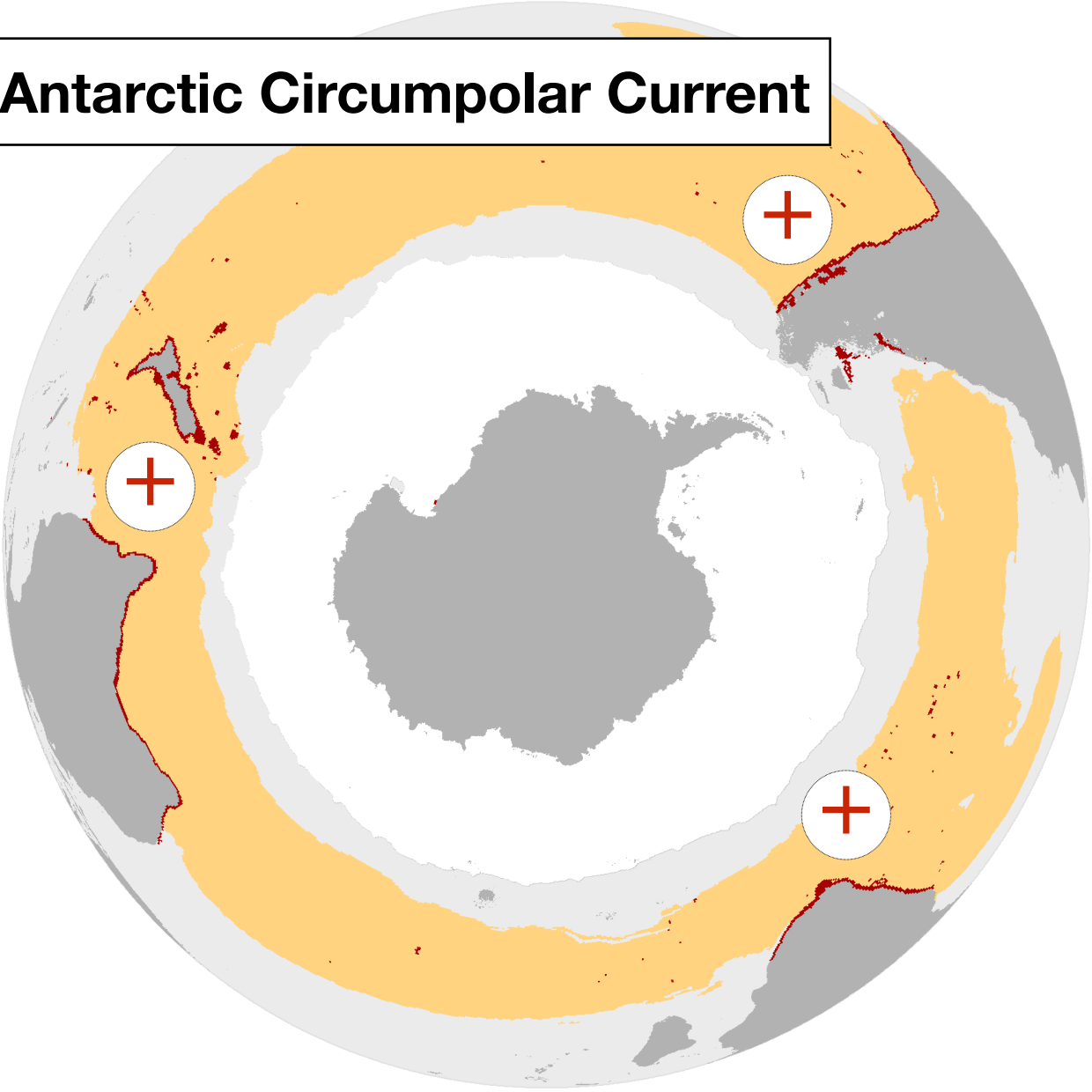
Past distribution [LGM]

Vancouver Island to
Baja California

Peru, southwards



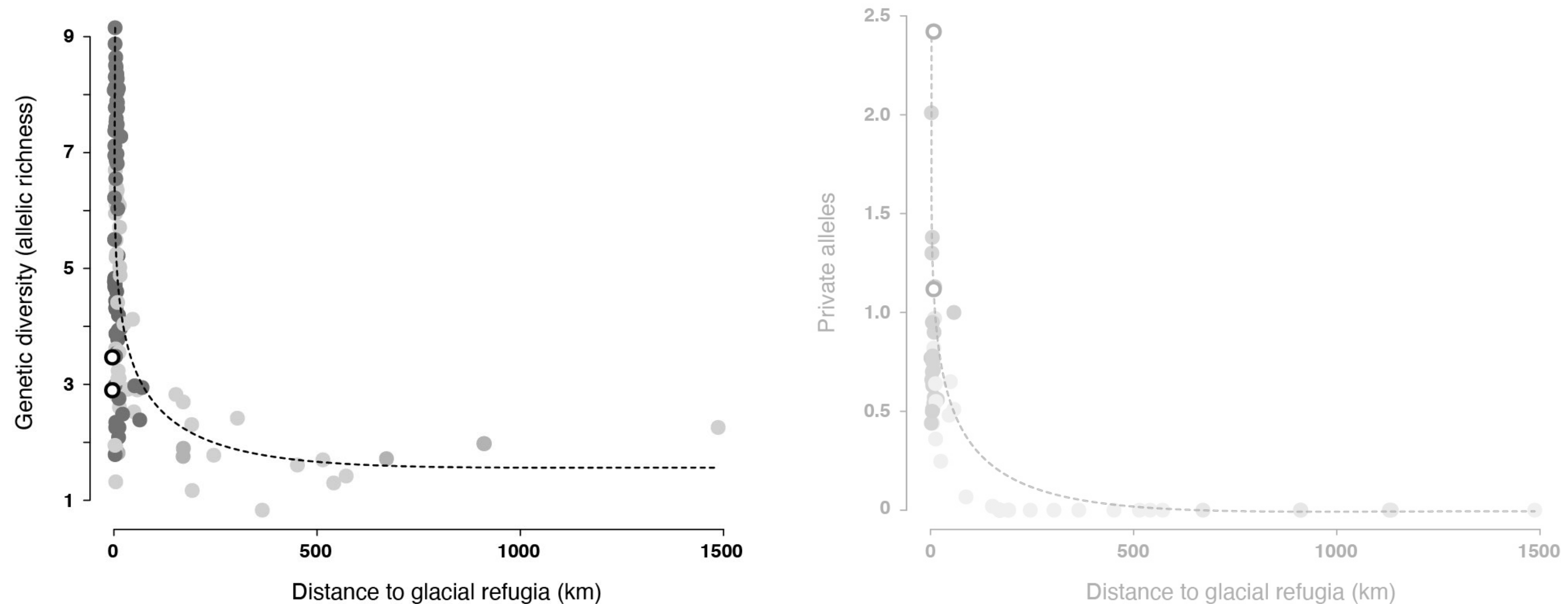
Antarctic Circumpolar Current



Comparing time periods

Multiple climatic refugia, where the species could have persisted.

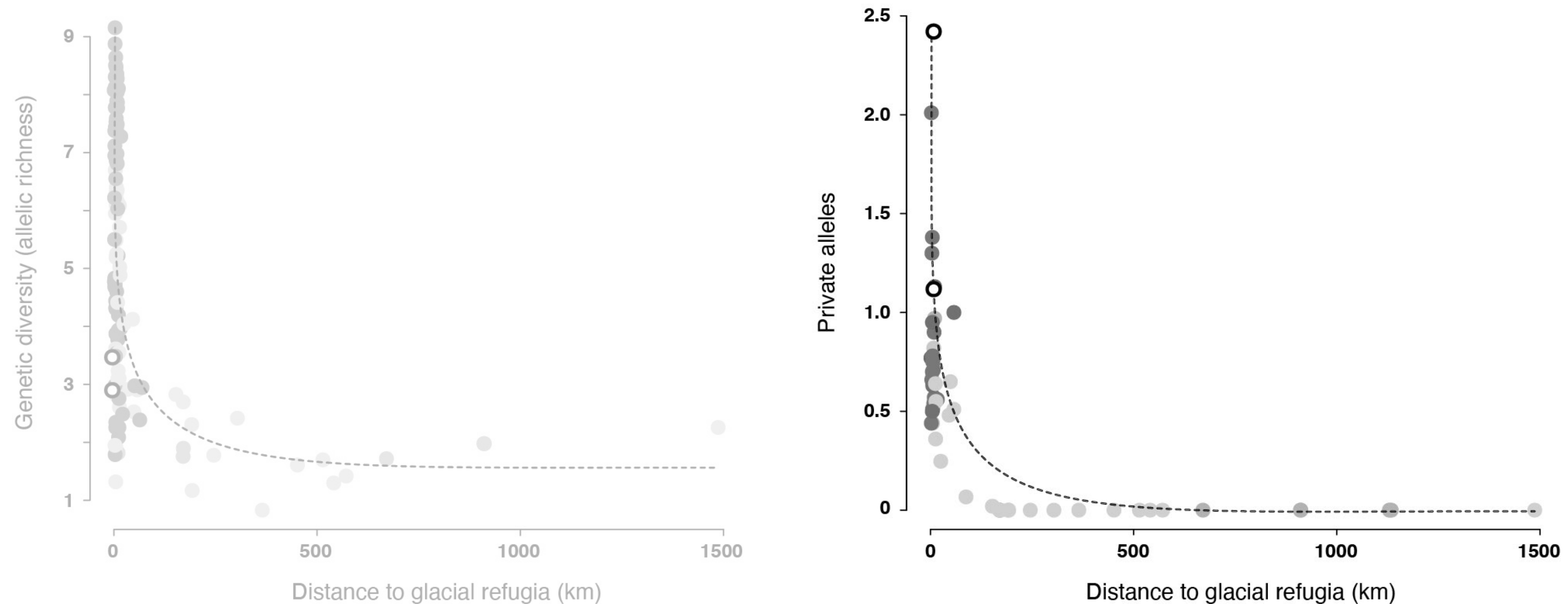
Comparison of genetic diversity vs. distance to climatic refugia (both hemispheres)



Diversity systematically lower outside refugia (regions of persistence).

Diversity up to 3-fold lower at distances higher than 75 km of refugia.

Comparison of genetic diversity vs. distance to climatic refugia (both hemispheres)



Diversity systematically lower outside refugia (regions of persistence).

Endemic alleles (unique of populations) were **<1 at distances higher than 50 km of refugia**, and zero at 150 km of refugia (50 populations with no endemic diversity).

Genetic diversity levels can be shaped by past climate oscillations
(first model with replication across the hemispheres).

Higher latitudes

Post-glacial expansions left populations with reduced and homogenous diversity.

Lower latitudes

Range shifts structured **long-term refugia, with populations harboring higher and unique diversity.**

Conservation & Management

Identifying and Protecting Vulnerable Habitats



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Contents lists available at [ScienceDirect](#)

Marine Environmental Research

journal homepage: <http://www.elsevier.com/locate/marenvrev>



Environmental drivers of rhodolith beds and epiphytes community along the South Western Atlantic coast

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Macroalgae
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Light
Nitrate

ABSTRACT

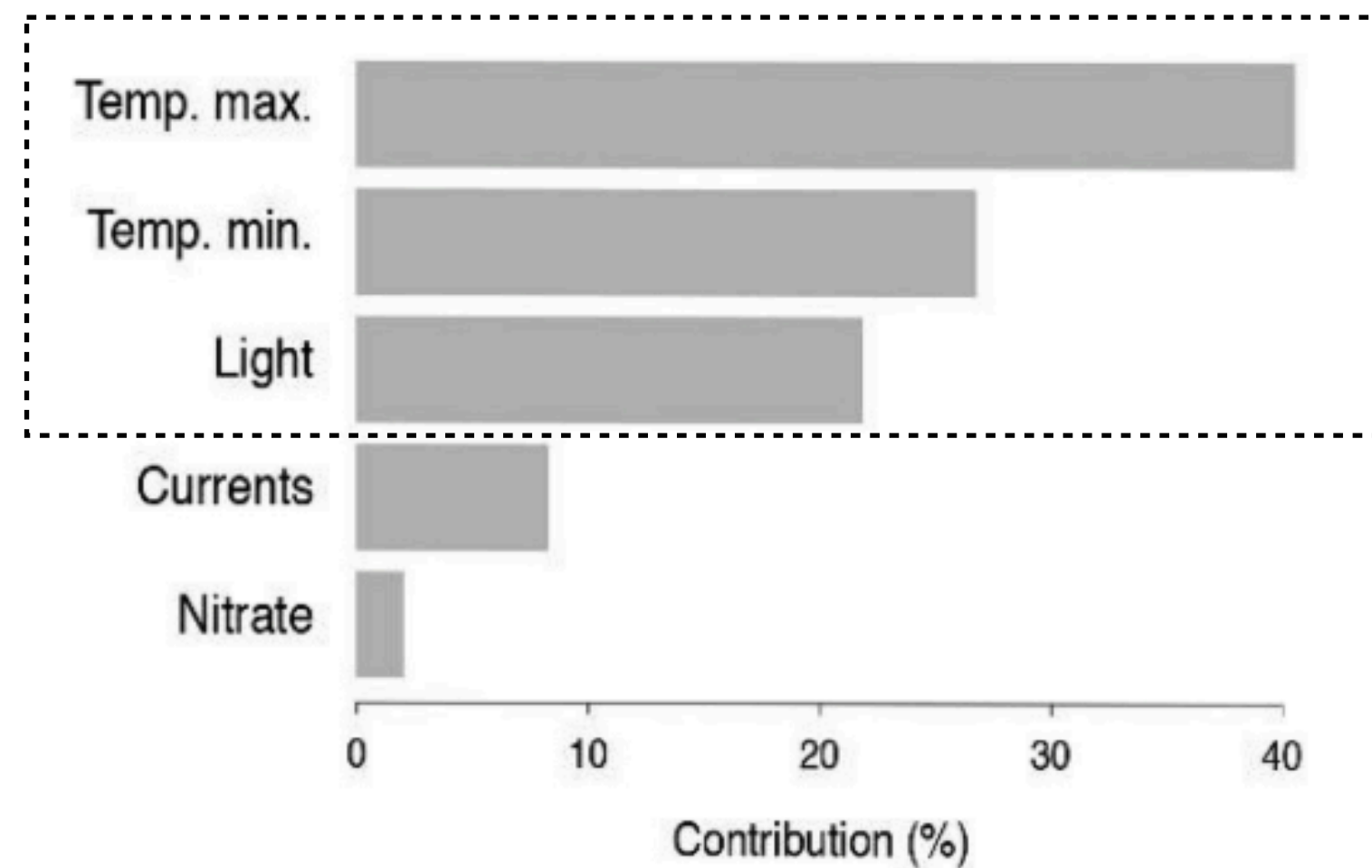
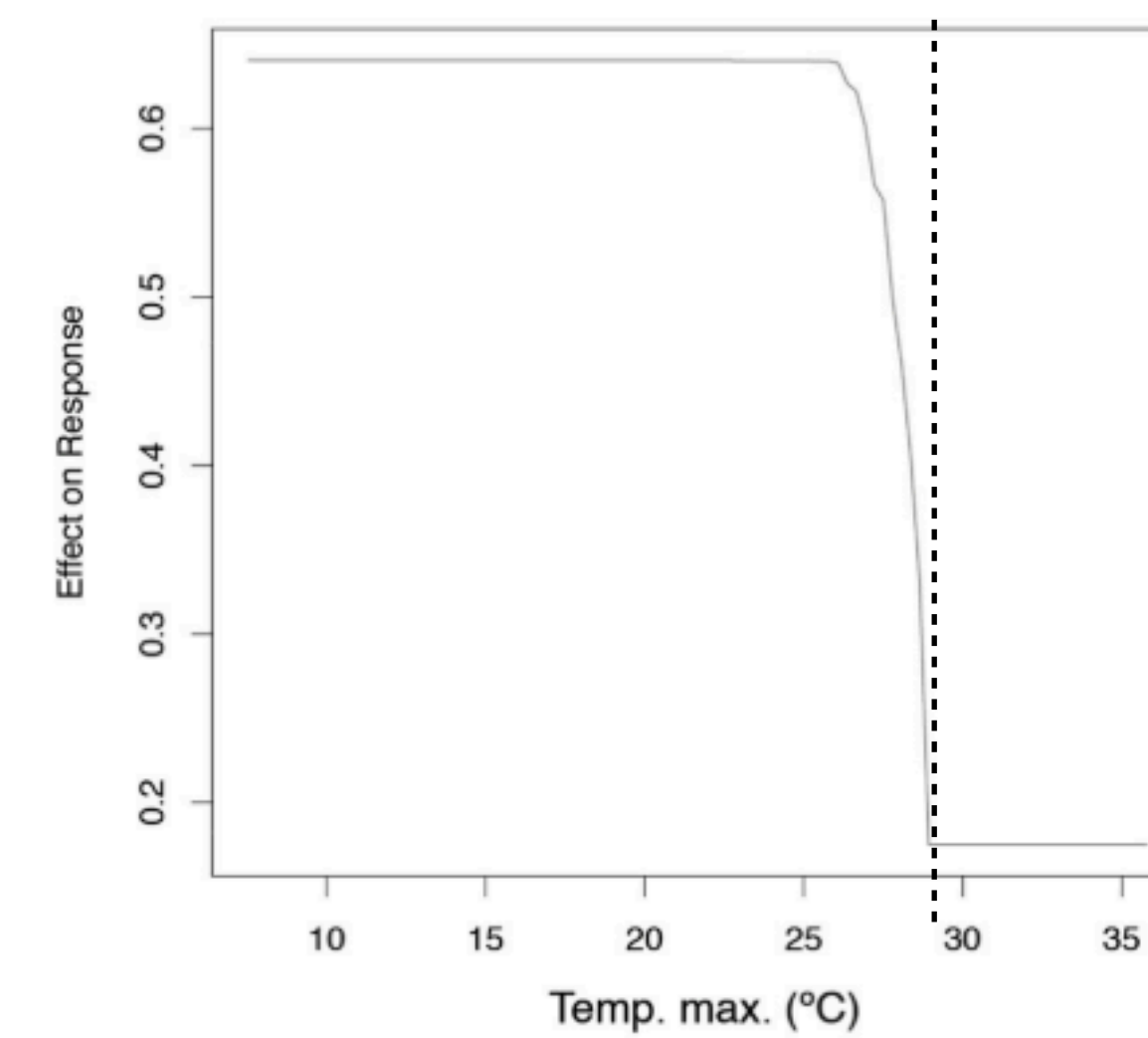
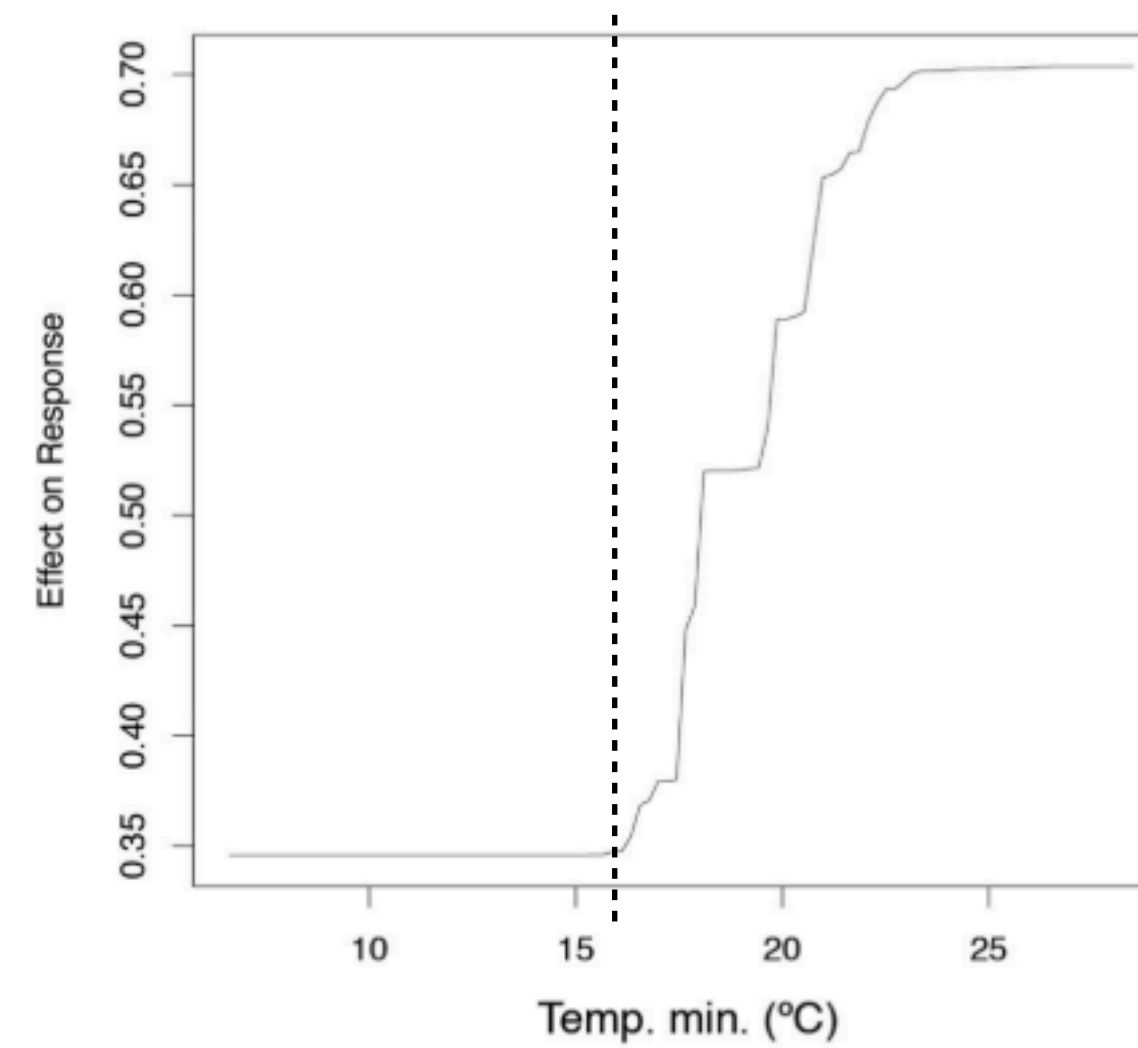
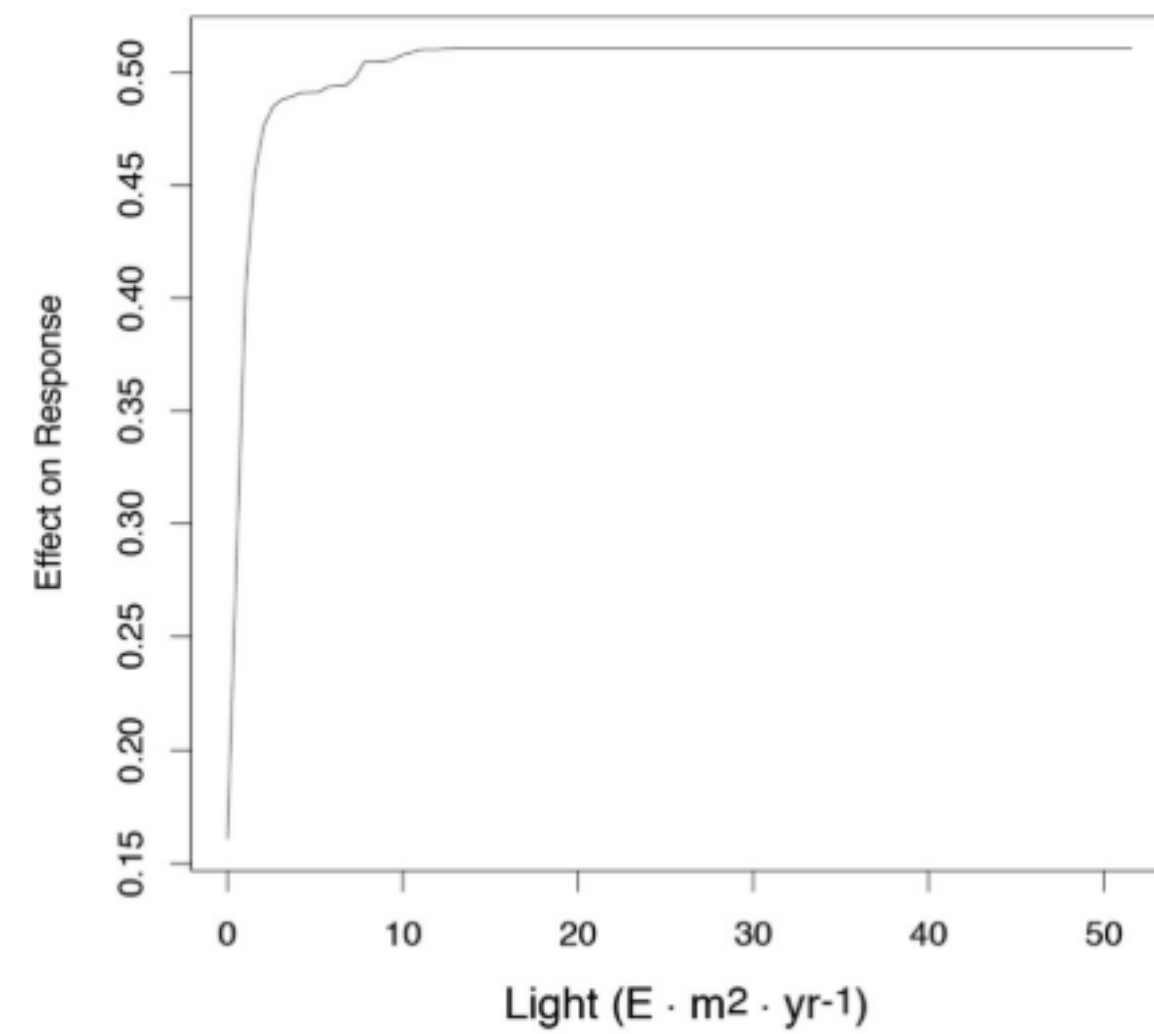
Environmental conditions shape the occurrence and abundance of habitat-building organisms at global scales. Rhodolith beds structure important hard substrate habitats for a large number of marine benthic organisms. These organisms can benefit local biodiversity levels, but also compete with rhodoliths for essential resources. Therefore, understanding the factors shaping the distribution of rhodoliths and their associated communities along entire distributional ranges is of much relevance for conservational biology, particularly in the scope of future environmental changes. Here we predict suitable habitat areas and identify the main environmental drivers of rhodoliths' variability and of associated epiphytes along a large-scale latitudinal gradient. Occurrence and abundance data were collected throughout the South-western Atlantic coast (SWA) and modelled against high resolution environmental predictors extracted from Bio-Oracle. The main drivers for rhodolith occurrence



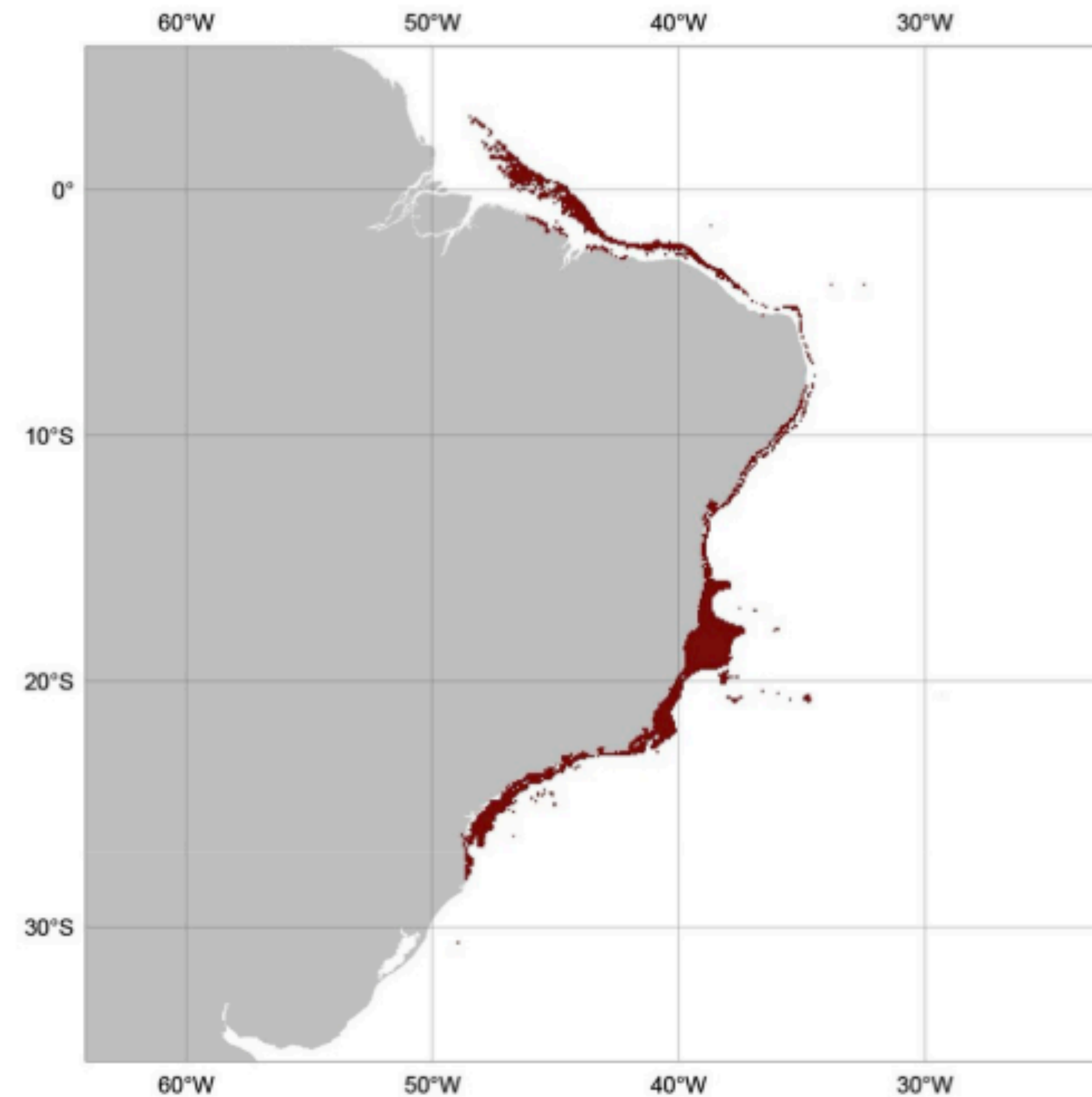
Rationale: Rhodolith beds structure important habitats for a large number of organisms. However, their **deep and cryptic occurrence hinders knowledge about where they are, and about the factors shaping their distribution.**

Goal: identify the distribution drivers and patterns of rhodoliths.

Methods: occurrence records collected throughout the South-western Atlantic coast and modeled against environmental predictors extracted from Bio-Oracle.



Results: main drivers for rhodolith distribution are light availability and temperature, with clear ecological responses supported by additional empirical studies.



Results: the Brazilian tropical-temperate region provides broad suitable areas for rhodoliths, decreasing toward extreme warm and cold regions.

Identifying and Protecting Vulnerable Habitats

SCIENTIFIC REPORTS

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Overlooked habitat of a vulnerable gorgonian revealed in the Mediterranean and Eastern Atlantic by ecological niche modelling

Received: 21 December 2015

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Published: 14 November 2016

Joana Boavida^{*,†}, Jorge Assis^{*}, Inga Silva^{*} & Ester A. Serrão

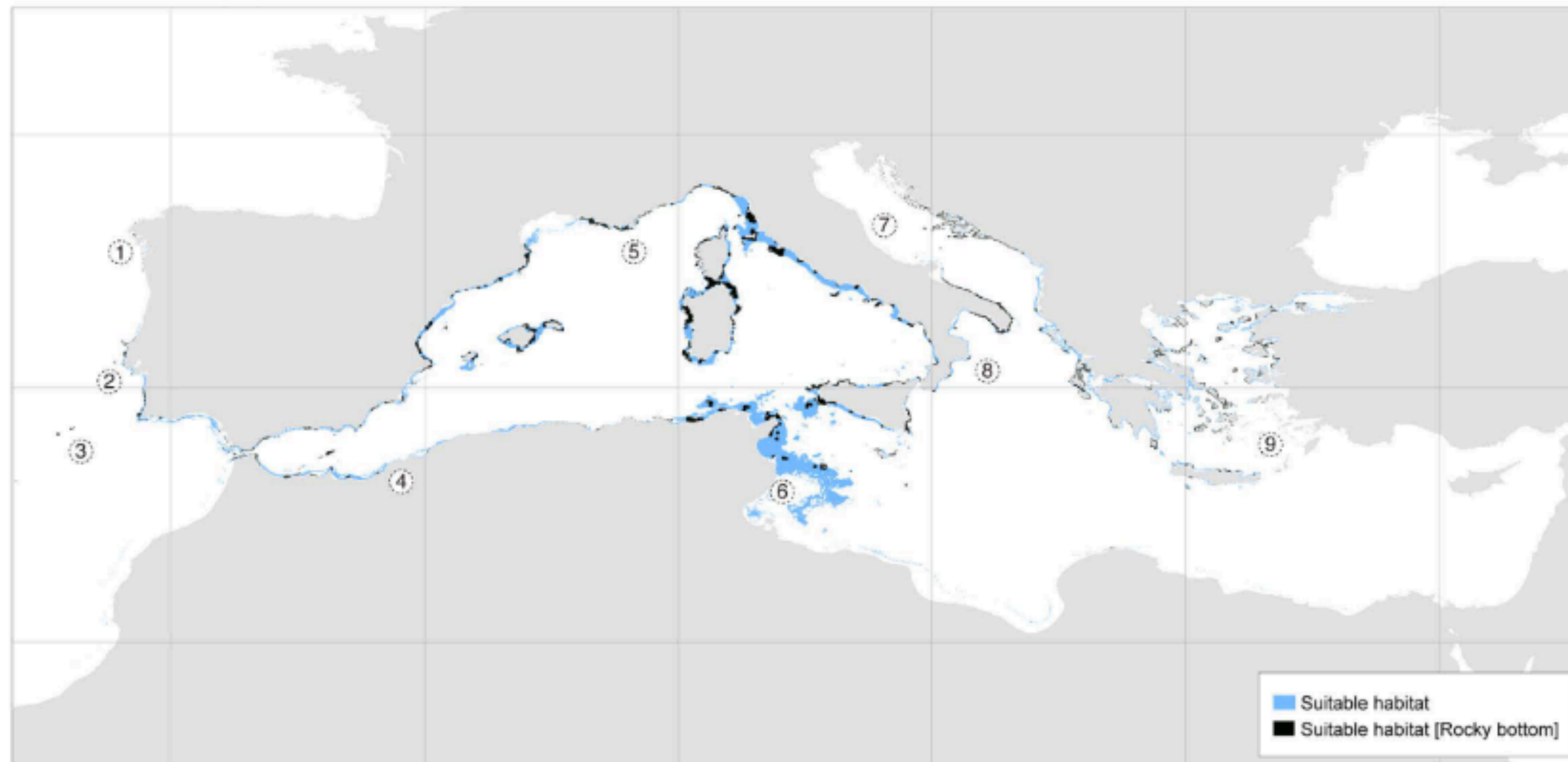
Factors shaping the distribution of mesophotic octocorals (30–200 m depth) remain poorly understood, potentially leaving overlooked coral areas, particularly near their bathymetric and geographic distributional limits. Yet, detailed knowledge about habitat requirements is crucial for conservation of sensitive gorgonians. Here we use Ecological Niche Modelling (ENM) relating thirteen environmental predictors and a highly comprehensive presence dataset, enhanced by SCUBA diving surveys, to investigate the suitable habitat of an important structuring species, *Paramuricea clavata*, throughout its distribution (Mediterranean and adjacent Atlantic). Models showed that temperature (11.5–25.5 °C) and slope are the most important predictors carving the niche of *P. clavata*. Prediction throughout the full distribution (TSS 0.9) included known locations of *P. clavata* alongside with previously unknown or unreported sites along the coast of Portugal and Africa, including seamounts. These predictions increase the understanding of the potential distribution for the northern Mediterranean and indicate



Rationale: corals remain poorly understood, potentially leaving overlooked distributions, particularly in non-surveilled areas (e.g., deep inaccessible rocky reefs). Knowledge about their distributions is crucial for conservation.

Goal: investigate the distribution of suitable habitats of an important structuring species (*Paramuricea sp.*) to inform conservation.

Methods: occurrence records collected throughout the entire Mediterranean and Atlantic distribution and modeled against environmental predictors extracted from Bio-Oracle.



Results: predictive maps increased our **understanding in areas of the deep northern Mediterranean (> 150m)**, and in **previously unknown habitats along Algeria, Alboran Sea and adjacent Atlantic**; encouraged deep exploration (NGS) to confirm its existence.

Evaluate new areas for invasive species

Science of the Total Environment 700 (2020) 134692



Contents lists available at [ScienceDirect](#)

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv



How experimental physiology and ecological niche modelling can inform the management of marine bioinvasions?

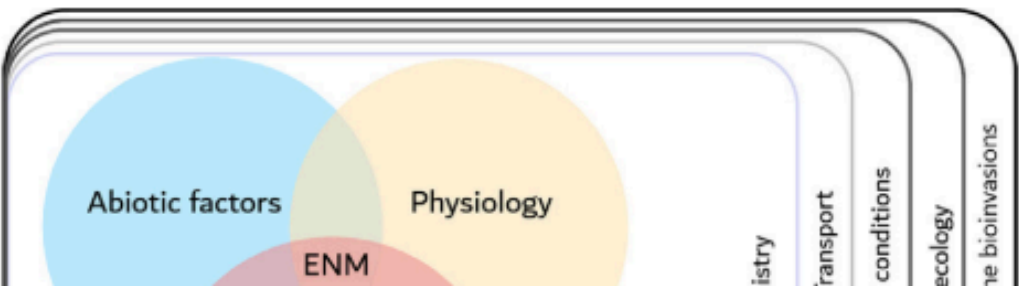
Gabrielle Koerich^{a,c,*}, Jorge Assis^b, Giulia Burle Costa^{a,c}, Marina Nasri Sissini^{a,c}, Ester A. Serrão^b, Leonardo Rubi Rörig^a, Jason M. Hall-Spencer^d, José Bonomi Barufi^a, Paulo Antunes Horta^{a,*}

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H I G H L I G H T S

- An Ecological Niche Model was built to explore the distribution of an invasive alga.
- Model results were crosschecked with ecophysiological evaluations.
- *G. turuturu* has higher habitat suitability in coastal areas.

G R A P H I C A L A B S T R A C T

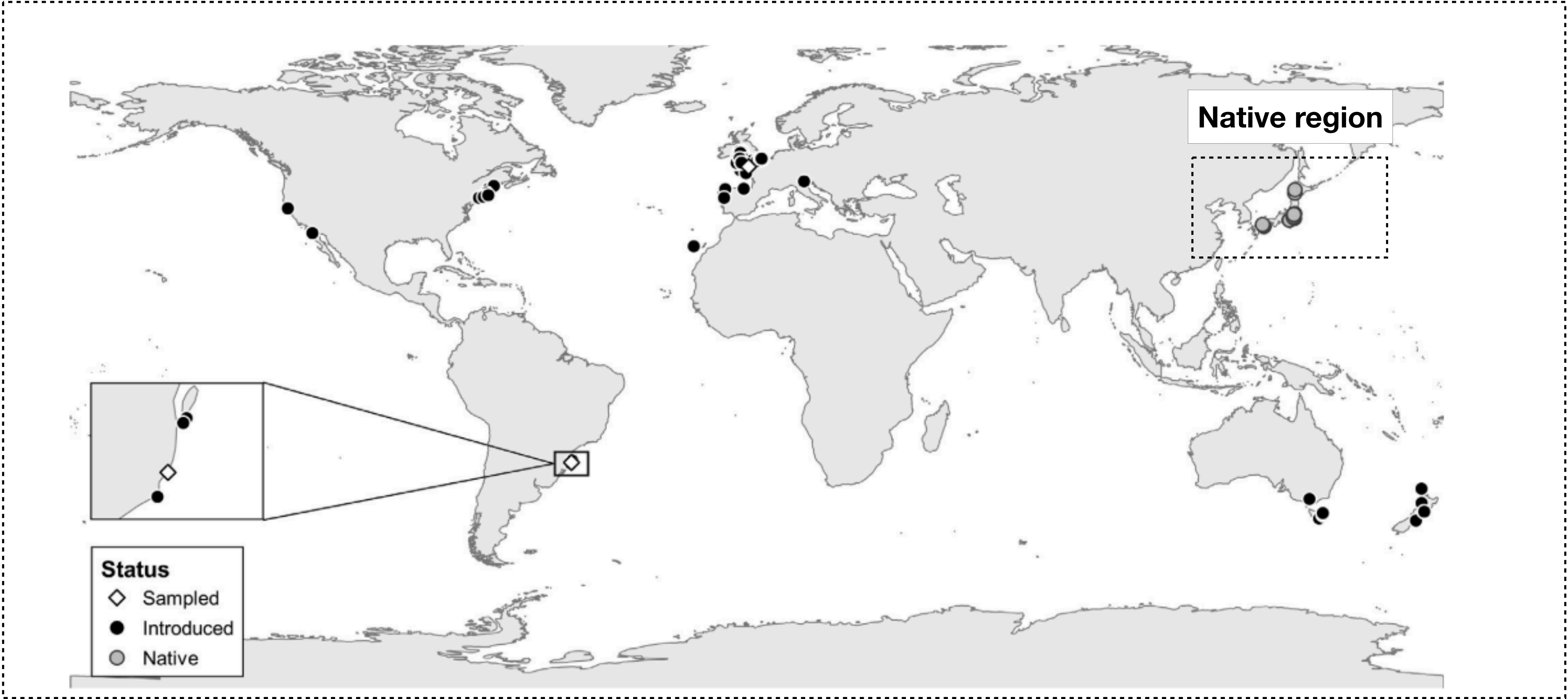


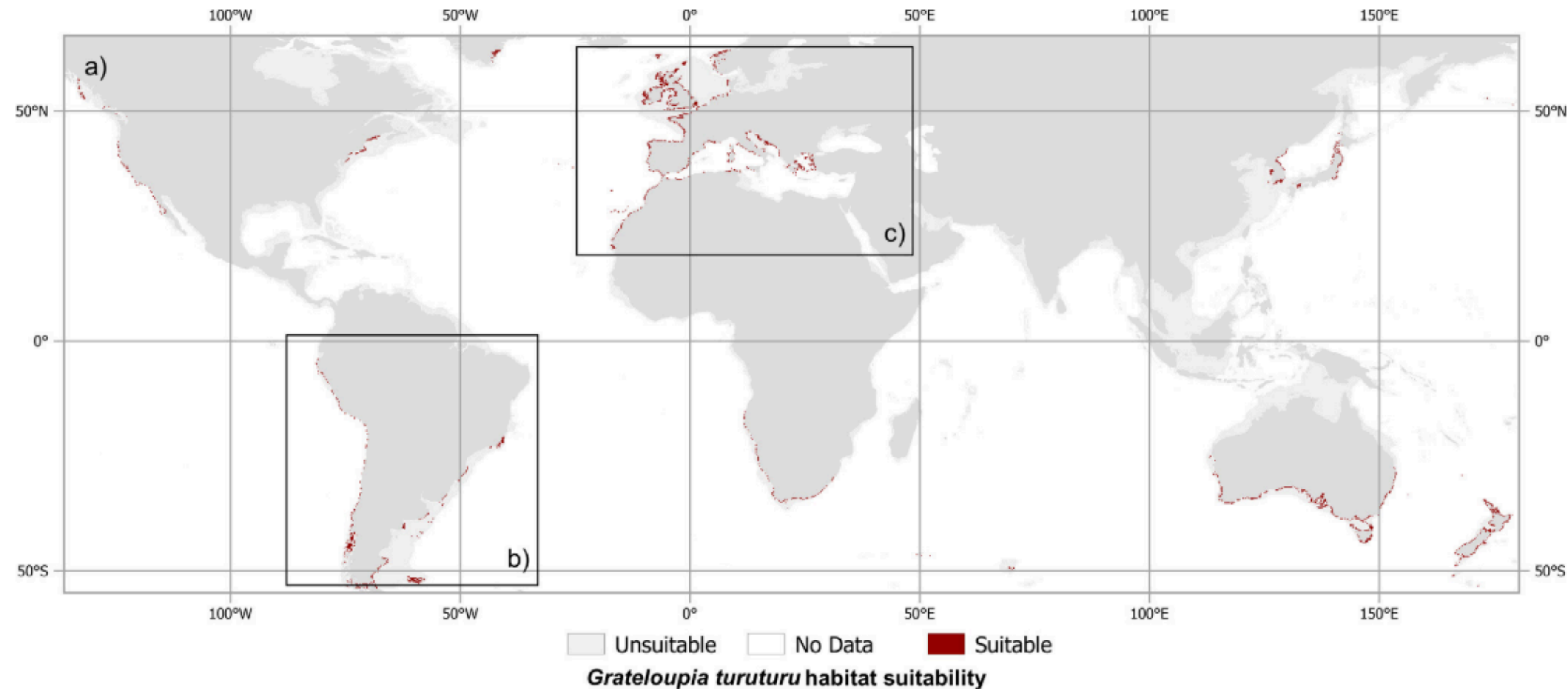


Rationale: information on **habitats of species with high invasive potential is crucial for management decisions** aiming to prevent their arrival and spread. *Grateloupia turuturu*, is one of the most harmful species, capable of causing biodiversity losses.

Goal: infer the potential global distribution to **assess potential areas of introductions**.

Methods: records collected by field sampling and the available literature, and modeled with high resolution environmental predictors extracted from Bio-Oracle.



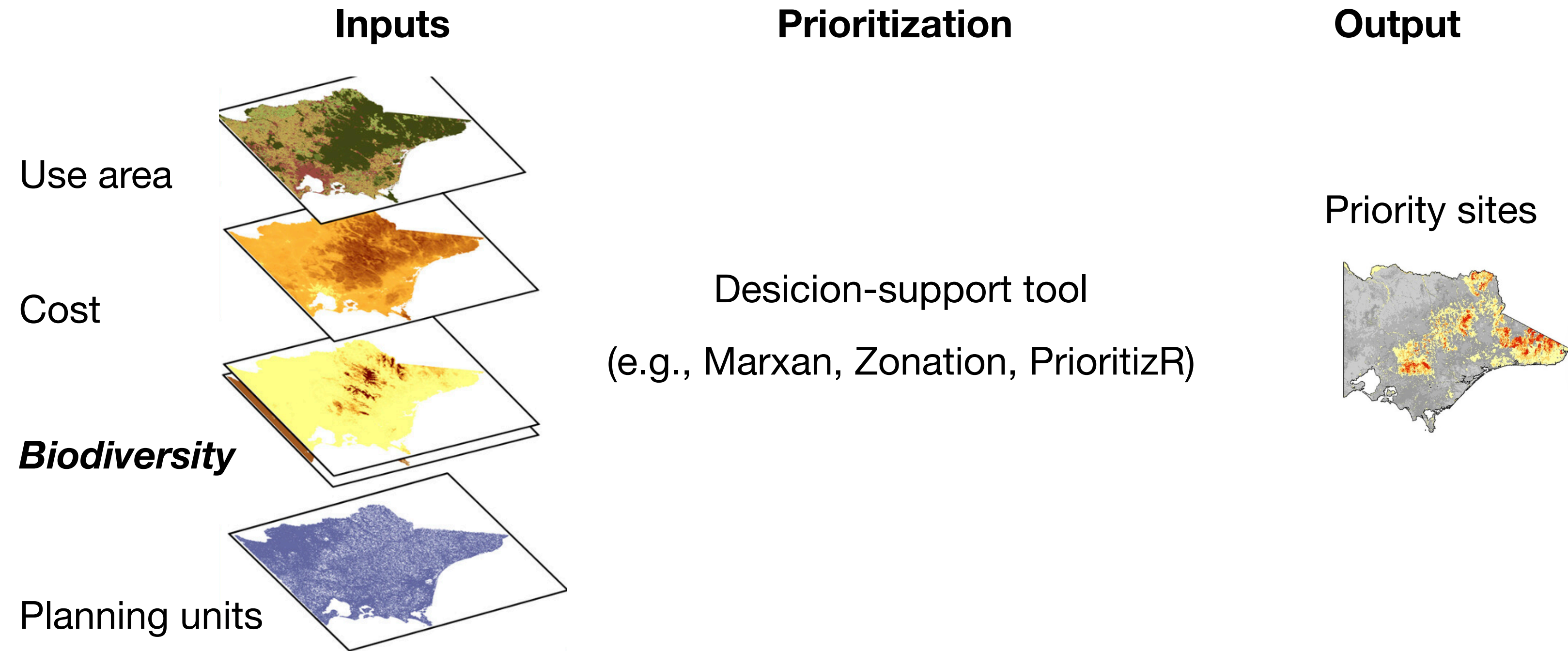


Results: **high suitability in temperate and warm regions** around the world, with **focus on areas where this species still doesn't occur**. Management initiatives can mitigate anthropogenic transport and promote eradication, with **focus in the South America, South Africa, Southern and Western Australia, and New Zealand**.

SDM maps are rarely the final product.

They are often the primary **input layers for spatial optimization.**

Marine Spatial Planning & Decision Support



Decision-Support Tools: Ingest SDM suitability maps alongside socioeconomic data (e.g., shipping routes, fishing zones) to **mathematically design optimal** protection networks.

FAIR Data and Reproducible Pipelines

Findable, Accessible, Interoperable, and Reusable

Making research more effective: Applying FAIR principles ensures that the data, code, and outputs associated with SDM studies are managed for maximum reuse and impact.

Findable: can others (and machines) find your data and models?

Accessible: once found, how can the data/model be accessed?

Interoperable: can the data be integrated and used with other data or workflows?

Reusable: can the data/model be effectively used in future studies?

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DATA ARTICLE

Global Ecology
and Biogeography

A Journal of
Macroecology

WILEY

Bio-ORACLE v3.0. Pushing marine data layers to the CMIP6 Earth System Models of climate change research

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Lennert Schepers³  | Lidiane Gouvêa¹  | Eliza Fragkopoulou¹  | Frederic Leclercq³  |
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Abstract

Motivation: Impacts of climate change on marine biodiversity are often projected

Data in Brief 53 (2024) 110200



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Data Article

A global dataset of demosp...

Data in Brief 48 (2023) 109223



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journal homepage: [www.elsevier.com/locate/dib](#)

Data Article

A dataset of cold-water coral distribution

[www.nature.com/scientificdata](#)

SCIENTIFIC DATA

OPEN

DATA DESCRIPTOR

A fine-tuned global distribution dataset of marine forests

Jorge Assis^{1,4}, Eliza Fragkopoulou^{1,4}, Duarte Frade¹, João Neiva¹, André Oliveira¹, David Abecasis¹, Sylvain Faugeton^{2,3} & Ester A. Serrão¹

Species distribution records are a prerequisite to follow climate-induced range shifts across space and time. However, synthesizing information from various sources such as peer-reviewed literature, herbaria, digital repositories and citizen science initiatives is not only costly and time consuming, but also challenging, as data may contain thematic and taxonomic errors and generally lack standardized



Check for updates

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O

A B S T R A C T

Species distribution data are key for monitoring present and future biodiversity patterns and informing conservation and management strategies. Large biodiversity information facilities often contain spatial and taxonomic errors that reduce the quality of the provided data. Moreover, datasets are frequently shared in varying formats, inhibiting proper integration and interoperability. Here, we provide a quality-controlled dataset of the diversity and distribution of cold-water coral

Data in Brief 52 (2024) 110023



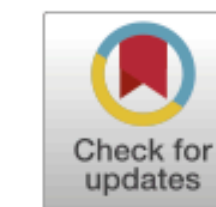
Contents lists available at ScienceDirect

Data in Brief

journal homepage: www.elsevier.com/locate/dib

Data Article

Range map data of marine ecosystem structuring species under global climate change



Lidiane Gouvêa^{a,*}, Eliza Fragkopoulou^a, T rence Legrand^a,
Ester A. Serr o^a, Jorge Assis^{a,b,*}

^a Centre of Marine Sciences, University of Algarve, Faro, Portugal

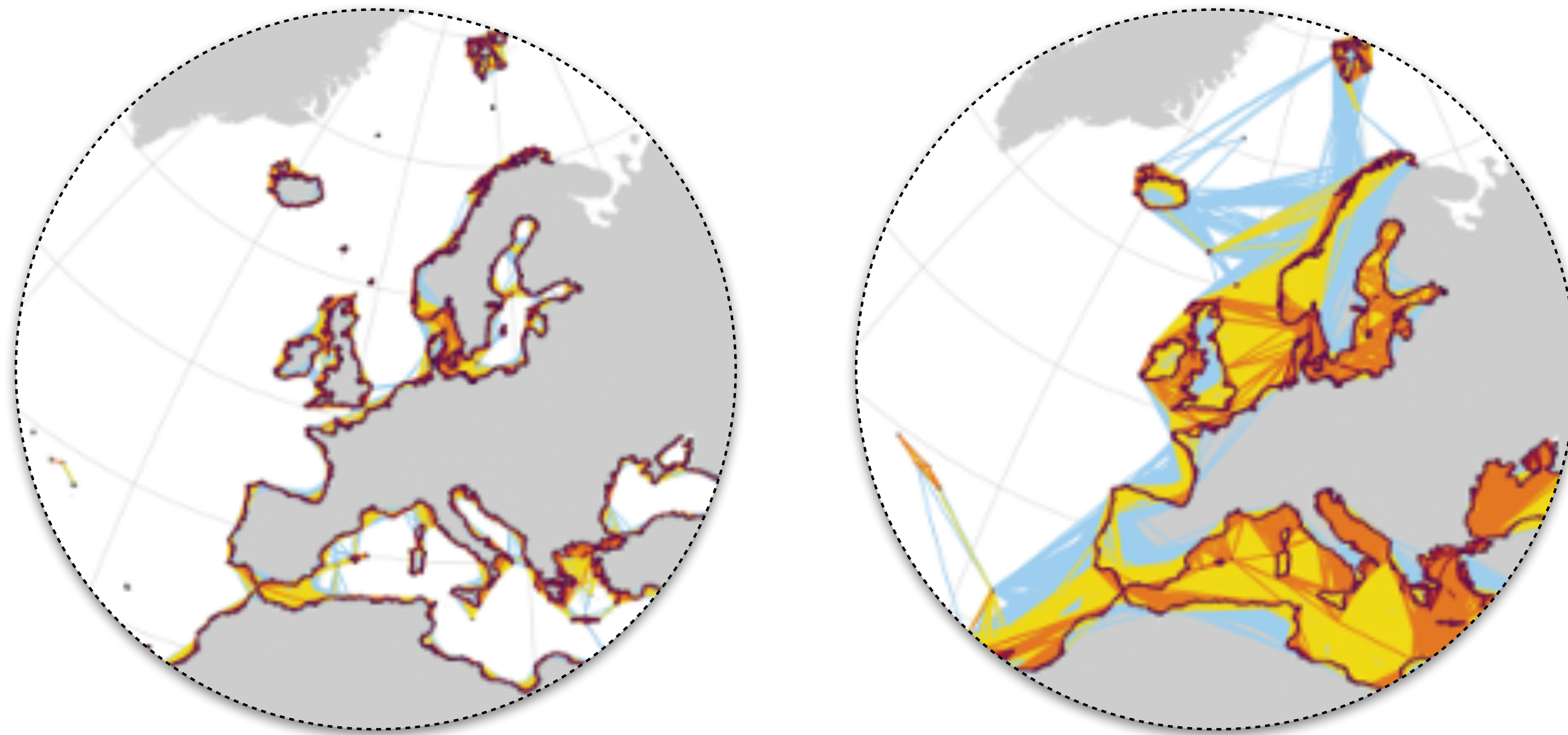
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ABSTRACT

Data on contemporary and future geographical distributions of marine species are crucial for guiding conservation and



github.com/jorgeassis/coastalNet

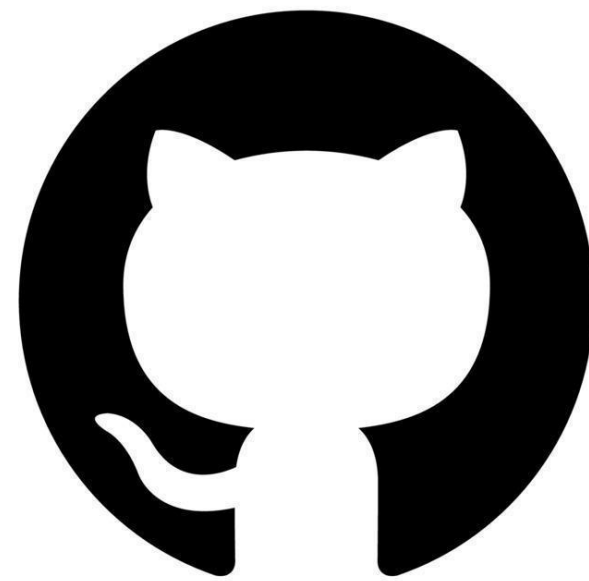
A dataset of oceanographic connectivity estimates along the world's coastlines derived from a biophysical model that assimilates 21-years of ocean current data.



Open access repositories

Online open access repositories allow researchers to **preserve and share their research outputs, including figures, datasets, images, and videos.**

E.g., Figshare and Zenodo are free to upload content and free to access, in adherence to the principle of open data.



GitHub

Git is an essential tool for managing complex analytical workflows like SDM, promoting reproducibility, collaboration, and saving us from losing work or getting lost in different versions of your scripts.

Git allows to access the exact version of the code used to generate specific results or maps, supporting transparency and validation (complements FAIR Principles).

References

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Carvalho et al., 2020: <https://doi.org/10.1016/j.marenvres.2019.104827>

Boavida et al., 2016: <https://doi.org/10.1038/srep36460>

Martins et al., 2021: <https://doi.org/10.1111/geb.13327>

Koerich et al., 2020: <https://doi.org/10.1016/j.scitotenv.2019.134692>

Assis et al., 2023: <https://doi.org/10.1038/s41598-023-38944-7>